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An empirical study of emoji usage on Twitter in linguistic and national contexts

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ABSTRACT

Emojis or 'picture characters' have become ubiquitous in modern-day digital communication, including social media sharing and smartphone texting. Despite this ubiquity, many questions remain about their usage, especially with respect to global variations in language and country. These questions are important, in part because they reveal how people communicate digitally on social platforms, but also because they provide a lens through which different regions and cultures can be studied. In this paper, we conduct a principled, quantitative study to understand emoji usage in terms of linguistic and country correlates. Our study involves 30 languages and countries each, and is conducted over tens of millions of tweets collected from the Twitter decahose over an entire month. Drawing on both statistical measures and information theory, our results reveal that, not only does emoji usage have strong dependencies at both the language and country level, but that some languages and countries are much more constrained in the diversity of their emoji usage. However, we also discover that the 'popularity' of emojis, both globally and within the context of a given language, follows a robust and invariant trend that emerges fairly quickly (over just a day's worth of data) and cannot be explained either by a power-law or Heap's law-like distribution.

1. Introduction

Emojis are pictographs that have become common units of expression, including for anger (💢), joy (😊), celebration (🎉), and similarly evocative expressions, in digital communication, including text messages sent through smartphones and social media platforms such as Twitter [1–3]. In a testament to their ubiquity as a visual language [4], Oxford Dictionaries declared the emoji 😂 to be the 2015 'word of the year', the first time an emoji had been granted this honor [5]. Because emojis have a distinctively social nature and are arguably *extra-linguistic* in origin, in that they are not part of the standard lexicon of any natural language, they have a special place in digital communication, representing 'shared culture' at multiple levels of granularity [6]. The 'smiley face' emoji (😀), for example, is widely used throughout the world to express a sense of joy or friendliness. Other emojis, such as 🐼 and 🐶, embody more regional cultures, settings and locales [7]. Emoji usage can also vary with the type of discourse in question. Expressions of love and tenderness tend to be associated with a certain set of emojis, as do sporting events (e.g., ⚽ and 🎂), holidays and birthdays (🎂).

An example of local 'shared culture' can be seen to manifest in Fig. 1, which covers a spatially large, but culturally and linguistically local region (the US state of California and some surrounding states). We find that there is high density and diversity of emoji usage along the coastline and also along the borders of states, with emoji usage seeming to recede as we move away from the coastline. This is not surprising since most of California's population is more concentrated along the coast, including coastal cities such as Los Angeles and San Diego in Southern California and the Bay Area (San Francisco, Palo Alto and others) in Northern California. The emojis not only reflect population but geography. Even more interestingly, the emojis clearly reflect culture e.g., the 🎃 emoji in several spots indicates that Halloween is near (in fact, further research showed that Halloween took place on the very last day covered in our corpus), and it is well established that Halloween is a popular holiday in the US. For reference, a non-US example (the south-eastern coast in Australia) is visualized in Fig. 2. While there are a number of similarities (heavy presence of 'smiley' emojis in both), some differences are also observed. Compared to California, there are fewer 🎃 in Australia than in California, and certain sports-related emojis (like the rugby) are entirely absent from Fig. 2.

However, these observations are only qualitative. While subjective studies, qualitative analyses and selective observations such as in the localized context of Fig. 1 can provide interesting clues to emoji usage, large-scale quantitative studies over a sufficiently broad spatial and

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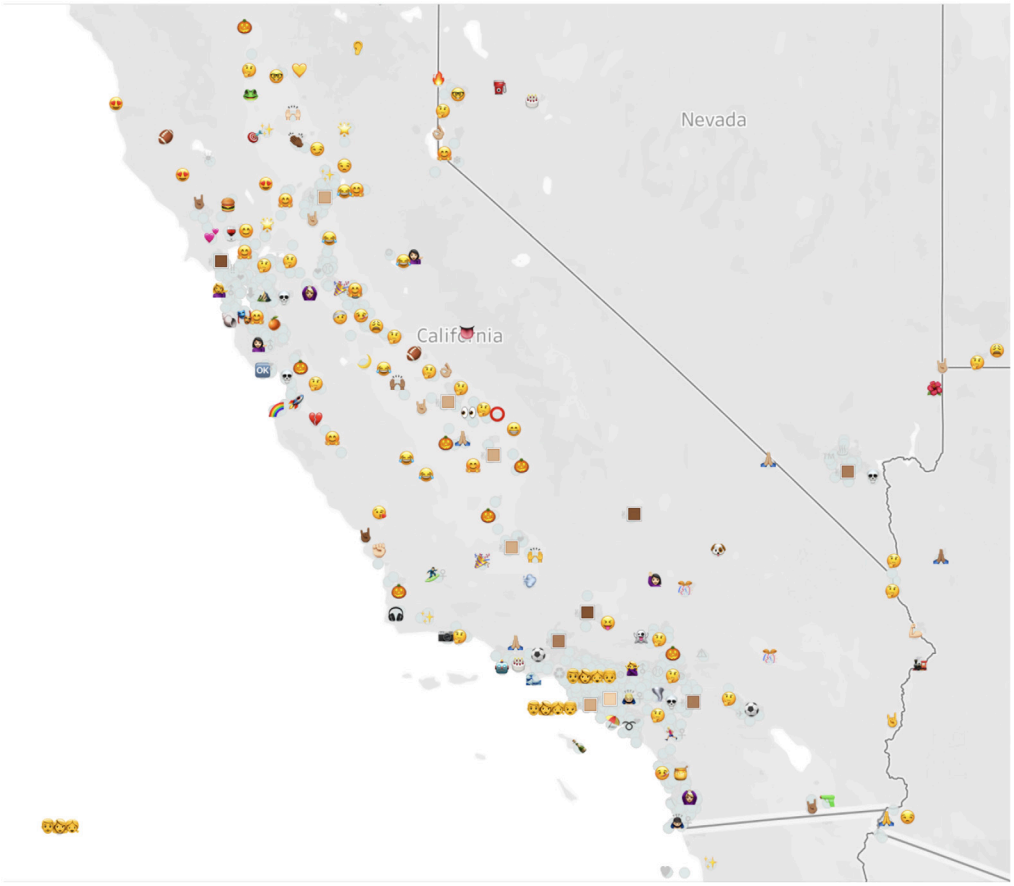


Fig. 1. A visualization of the most popular emojis in a narrow geographical region (the US state of California and its borders with Nevada, Arizona and Mexico) in our corpus. The emojis in the 'ocean' are likely from a cruise ship.

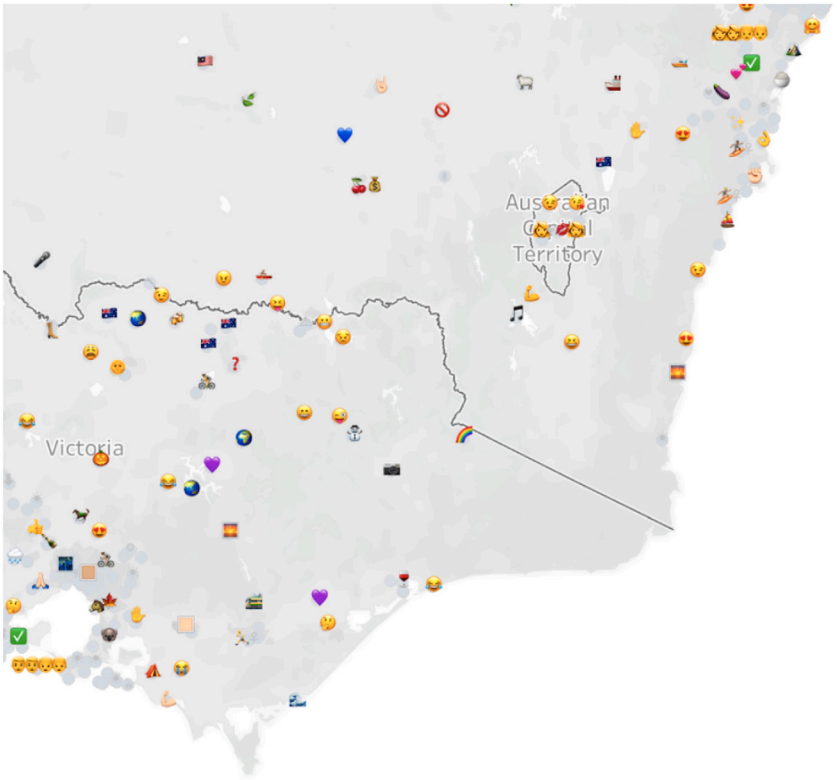


Fig. 2. A visualization of the most popular emojis around the south-eastern coast in Australia.

linguistic distribution have been rare, though a number of papers have attempted to study emojis in more limited settings (Section 2). It is not fully evident from these studies how emoji usage differs across languages and countries, and how the degree of difference could be measured rigorously.

Studying emojis in a global context (i.e., not just for a specific country, such as the United States, or a particular language, such as English) is also motivated by the advent of social media-based Natural Language Processing (NLP) applications in multi-lingual settings [8,9]. Such applications include sentiment analysis, multi-lingual chatbots, opinion mining and influencer marketing [10–13]. Many NLP applications today rely on deep learning methods, including representation learning [14]. Traditionally, representation learning has been limited to ordinary linguistic tokens, such as words and sentences [15]. Since emojis are an important part of social media communications, embedding them can yield a higher-quality signal for classification, as suggested in works such as [10] and [14] (see also related work in Section 2). However, the majority of current NLP research on emojis does not distinguish them on the basis of the linguistic or national context in which they are used. In this article, we attempt to quantitatively understand the dependencies between emoji usage (and popularity) and these linguistic-national contexts.

With these motivations in mind, we propose and conduct a set of detailed empirical studies with the emoji as our primary object of investigation. In particular, we study how emoji usage depends on both language and country using rigorous distributional and information-theoretic methodologies. The study is performed using a large amount of Twitter data collected globally from the 'decahose' (a random 10% of all publicly streaming tweets). Because we also have the country codes associated with the geolocation where the tweet was actually posted, we are able to measure linguistic and national dependencies of emoji usage without depending on automated tools that only indirectly infer location by using text-based features (often with less-than-perfect accuracy [16,17]).

Our findings in Section 3 reveal some interesting facts about emoji usage that, to the best of our knowledge, have not been reported before. Specific contributions and empirical highlights are enumerated below:

1. As a first contribution, we do a study on emoji *prevalence* (the extent to which emojis are used in tweets), and find that, while the prevalence follows a power-law distribution globally, not all languages or countries use all emojis equally (even after controlling for sample size). In other words, after controlling for size, there are differences between languages and countries in the numbers of unique emojis found in their tweets (Section 3.2).
2. We study emoji *popularity* (the relative degree to which emojis are used in tweets) and find that the popularity of emojis seems to follow a robust trend that is language-independent, and that emerges fairly quickly (in less than a day's coverage of data), remaining largely consistent thereafter. However, we also find that this trend obeys neither a standard power law nor a Heap's law-style distribution, although we hypothesize that it could still be studied as a *3-phase piecewise* power-law distribution (Section 3.3.1). To the best of our knowledge, this 3-phase fractal structure of emoji popularity has not been shown in prior work.
3. We do a detailed study on how emoji *usage* varies across languages by using an information-theoretic measure called the Kullback–Leibler (KL) Divergence (Section 3.3.2). This measure allows us to model emoji usage as a probability distribution and to compare *relative* emoji usage between two languages.
4. We repeat the study above for countries and find the results to be similar, but we also find that countries (and in particular, the Middle East countries) are more differentiated in their emoji usage than languages (Section 3.3.3).

In addition to the quantitative studies described above, we also supplement our analysis with qualitative studies that illustrate how emojis can be both universal and local. The remainder of this article is structured as follows. Section 2 provides a summary of related work, while Section 3 presents the experimental study. In Section 4, we summarize key empirical findings. Section 5 provides a discussion, including some qualitative results, and the article concludes (with some guidance on potential future research) in Section 6.

2. Related work

Emojis have become gradually popular as subjects of study (and also for related applications, such as sentiment analysis, of which three good examples include [10,14,18]), mirroring their popularity as a visual mode of communication for social media and mobile users [1–3,6]. For example, according to [1], the 'introduction of emojis is a potentially dramatic shift' in online discourse, and the specific question they address is whether individual users' use of emojis reduces the frequency of emoticons, which are different from emojis, as they are expressed using the standard characters found on a keyboard e.g.,): and :(. They use a causal inference approach to analyze this hypothesis. [2] instead refers to emojis as a 'new generation of emoticons' that was continuing to grow in popularity and usage in both mobile communications and social media. According to them, over 10 billion emojis were used on Twitter in the period 2013–2015. This suggests that studying emojis could become important to studying digital communication. [3] studies a slightly different issue, namely the roles of emojis in mobile phone notifications. They find that the emojis sentimental polarity is high on Twitter but the overall number of emojis is lower on Facebook. Their results allow them to draw conclusions in relation to the number, position, type and sentimental value of emojis. Finally, Kimura-Thollander and Kumar [6] show that there is a possible cultural gap between user perceptions and the emoji standard at the time. Using participatory design, they address the gap by designing forty emojis and evaluate their acceptability compared to existing Japanese emojis using surveys. Our work is complementary to these more sociological and human–computer interaction studies.

Some of the papers cited above have relied on emojis for more accurate sentiment analysis [10,14,18], which has important applications in e-commerce and marketing (among others). For example, Chen et al. [10] propose a sentiment analysis method that 'embeds' the emojis in a vector space using an attention-based model. They are able to achieve significantly better results, as a consequence. Similarly, Eisner et al. [14] propose emoji2vec where they learn emoji embeddings from their description. As they discuss in their paper, emoji embeddings, along with word embeddings, could be useful for Natural Language Processing (NLP) applications that operate on social media inputs.

A more comprehensive treatment and study, especially on the symbolic representations of identity and collective behavior in Twitter, may be found in the dissertation by [19], which uses emoji use to study this important topic. One of their research questions is, in fact, a study of the differences in emoji use across users and documents in social media. Complementary to that study, which looked at individual users and their social identity (as well as individual communication style preferences) as the primary subject of focus, we look to measuring differences in emoji use by considering countries and languages as primary foci. Future work may even try to combine these two sets of results and methodologies to conduct a more 'mesoscopic' analysis.

A related study to the one in this article by Ljubešić and Fišer provides global analysis of emoji usage [20], but their analysis is orthogonal from ours in multiple respects, including the methodology and data collection. While the authors in [20] focused on the issue mainly through the lens of what they referred to as 'first world', 'second world', 'third world' and 'fourth world' nation clusters, our primary focus is understanding difference in emoji use by considering the influence of both language and country. Our study is therefore complementary to

theirs, and allows us to consider emoji use from yet another lens. Future work could even try to combine the results as they are at different levels of granularity and aggregation.

In another very recent work, [21] use emoji usage to study cultural differences through yet another lens, namely by comparing the usage across the East and the West. At least one of their research questions is similar to ours, namely 'how does the frequency of different emojis vary across the East and the West?' Our work is complementary but at a finer granularity as we use information theory to study the differences on a per-country and per-language basis. As [21] themselves point out, prior studies on emojis can be divided into three themes. The first of these concerns sentiment annotation, which we briefly mentioned above. The second is analyzing differences in emoji perception based on rendering, an interesting topic but one that is orthogonal both to their analysis and ours. The third theme is understanding similarities and differences in emoji usage across different populations. Both this work, as well as several others cited in this section (including [21]) fall squarely under this theme.

A recent article that is more extensive than ours in period of coverage is the work by Li et al. [22]. They perform an empirical analysis of emoji usage using a large-scale, cross-regional dataset; however, their primary object of investigation is emoji usage dependency on the Hofstede's Cultural Dimensions Model [23]. The study is complementary to this work, as it considers emoji usage from a different perspective. While some of our own findings also have a cultural dependency (since language and country are significant cultural correlates), we focus on language and country as variables of analysis. We also study emoji popularity, both globally and in specific linguistic contexts, which was not the focus in their study. Also, in part, they do automatic geolocation inference, which can lead to noise [16]. Our dataset is limited to those tweets where the metadata provides us with the user account's country code with certainty.

Beyond studies with broad range, a number of studies have looked at emojis in specific contexts. A good example is the work by Barbieri et al. [7], which considered patterns of Twitter emoji usage in Barcelona and Madrid and found that, while overall semantics of emojis are preserved within these regional constraints, interpretations can differ, leading to the conclusion (borne out by studies in [22] much later) that emojis can have different culture-dependent interpretations. In other related work, studying emojis from a network science perspective, including through the construction of co-occurrence networks, has been the focus of a few studies [24,25]; these studies provide a complementary lens through which to study emojis. Our own work primarily focuses on language and country correlates of emoji usage, rather than on co-occurrence networks.

The use of information theory for rigorously and quantitatively studying language and country dependencies on emoji usage is an important methodological contribution of this work. The use of information theory in social media has started taking hold (especially in application and evaluation-oriented contexts) [26–28], although it is still in its infancy. We refer the reader to the very recent work by Bagrow and Bollt for an information-theoretic approach to comparing networks [26]. To the best of our knowledge, such metrics have not been used in social media studies of this nature. However, information theory as we use it in this article has been popular in multiple other communities and applications including artificial intelligence and neural networks, physics and wireless communications [29–31].

Beyond computational studies, emoji usage has also been studied in media studies and sociology of late. For example, Alshenqeeti [4] questions whether emojis should really be considered to be a 'new' visual language for new generations by performing a socio-semiotic study. Riordan [32] considers the role of emojis as 'tools' for 'emotion work', specifically in the context of communicating affect of text messages. She found that emojis of objects communicate positive affect and framed her findings in the sociological theory of emotion work (in essence, suggesting that the time or effort involved in emoji usage helps maintain and even enhance social relations). Similarly, another study (based on emojis in the comments of a WeChat official account) found that emojis can be used to perform and reinforce a sense of 'playfulness' but that this belies a 'discursive conformity to social norms' in real life. Emojis were therefore found to straddle the tension between the openness of social media and the conservative nature of established social norms [33].

In yet other work, Hu et al. consider the intentions and sentiment effects of using emojis and employ detailed hypothesis testing [34]. All of these findings reinforce the points made in the introduction about the significance of emojis in modern-day digital communication. Our study proposes to add to this body of knowledge by understanding emoji usage more fully in the context of the language (in which the tweet is written) and the country from which the text was tweeted, using a novel information-theoretic methodology.

One aspect to note about our study is that its conclusions specifically hold only for users who enabled their geolocations. As described by Malik et al. [35], geotagged tweets can be associated with certain selection biases, including factors such as higher median income, being in an urban area, being along a coastal area, higher prevalence of young people and greater proportion of Asian, Black or Hispanic populations. Nevertheless, as they also cover in their paper, the availability of geotags provides a rich basis for study, and many authors have used these tweets for computational social science studies [36–38]. Our study draws on those precedents, although extending and comparing our results to non-geotagged tweets is an important avenue for future research.

3. Experimental study

3.1. Preliminaries: Material and methods

We conducted our experiments on tweets collected from the Twitter decahose through all of October, 2016 (29 days total,[2]). While the decahose contains a random, continuously streaming 10% of all Twitter data (which amounted to a sample of 50 million tweets per day at the time with global coverage), we filtered the dataset so that only tweets with geographic coordinates were retained[3] leading to a total of 19,562,132 tweets in our 29-day corpus. Since approximately 22.05% of these tweets contain at least 1 emoji[4] and we used the 30 countries and top 30 languages to avoid issues to data skew, the total number of tweets in our cleaned corpus (each of which has both a geolocation as well as one or more emojis) numbers 4,057,871. Since the tweets are global, multiple languages and countries are represented, although English continues to be the dominant language. Because we have only retained tweets with geographic coordinates, all results (starting from the descriptive statistics in the next section and also in the appendix) must be interpreted in this context. As we had earlier described in Section 2, the work by Malik et al. [35] and other researchers suggests that geotagged tweets can be associated with certain selection biases.

[2] Due to data collection errors, the stream for Oct. 14 and 23 could not be reliably retrieved; hence we did not use them in our experiments.

[3] Specifically, inside a tweet's metadata (represented as a dictionary), we look for a key called 'place', which itself references a key–value dictionary. Inside this dictionary, we check for non-null values for a 'coordinates' key. Twitter itself provides a 'country_code' associated with these coordinates, assuming they exist. We use this country code for associating the tweet with its originating country.

[4] We use the package https://github.com/carpedm20/emoji/ (Version 0.6.0, which has a corpus of a total of 3859 emojis, released on Aug 1, 2020) for isolating the emojis in the tweets for supporting the experiments in this article. In total, we extracted 1885 unique emojis across all tweets in the corpus under consideration. One reason that not all 3859 emojis were extracted is that this corpus is from 2016, when not all emojis in the latest corpus were available; furthermore, some emojis are generally used very rarely.

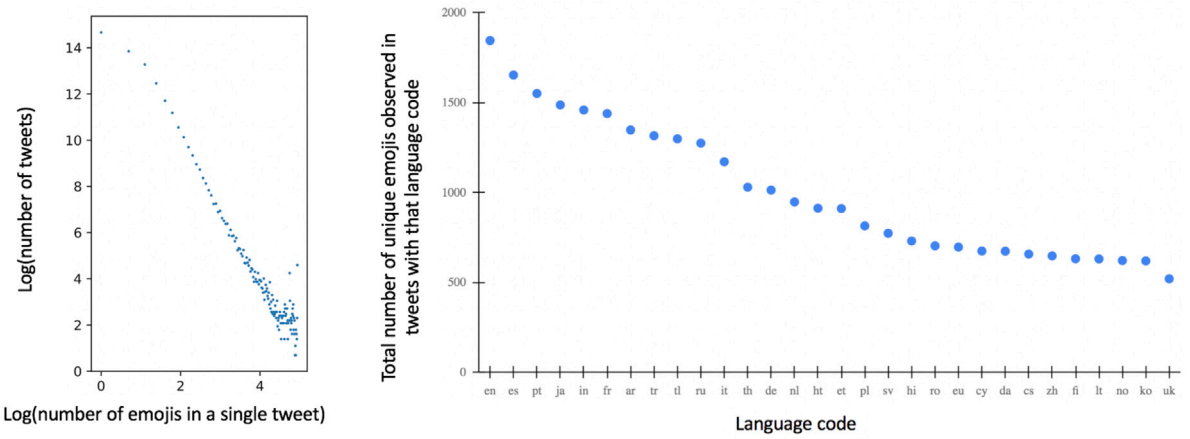


Fig. 3. A distributional overview of our data. (a) measures the frequency of the tweets (on the natural log scale) versus the number of (not necessarily unique) emoji occurrences within the tweet (also on the natural log scale); (b) measures the number of distinct emojis recorded in the corpus per language. We consider the top 30 languages in our corpus (rank-order plot), and have verified that each of them has at least 10,000 geotagged tweets in our corpus for the month of October, 2016.

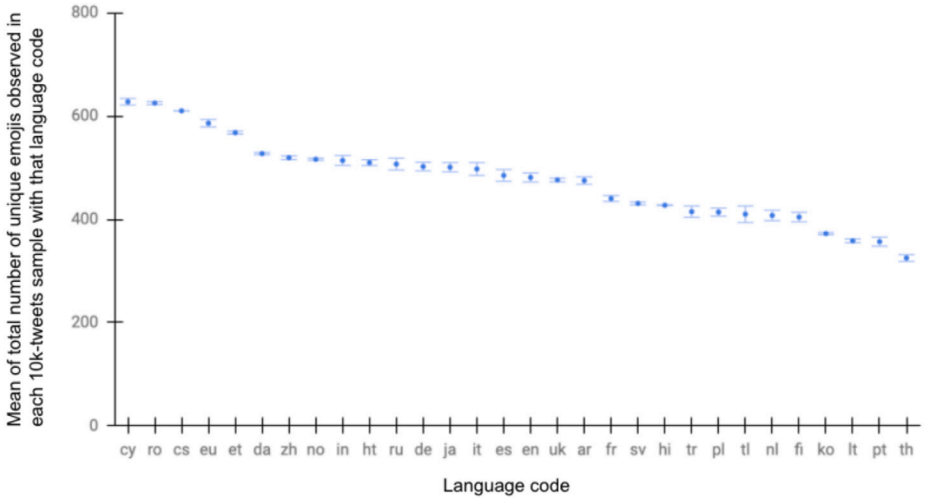


Fig. 4. The number of distinct emojis recorded in the corpus per language, for the top 30 languages in our corpus (rank-order plot). Unlike Fig. 3(b), we controlled for different numbers of tweets per language by randomly sampling 10,000 tweets per language. To control for the effects of randomness, we conduct 10 iterations, and report with error bars.

In addition, to verify the quality of some of the metadata, including the correctness of the language tag provided by Twitter, as well as whether the tweet is a 'spam' tweet or not, we conducted two manual studies. First, to verify that the majority of tweets are non-spam, we randomly sampled 300 tweets in English and manually checked the proportion of the 300 tweets that could be classified as spam. However, we only found 1 spam tweet among the 300, suggesting that only a minority of tweets are spam.

To check the correctness of Twitter's language tags, we sampled a total of 592 tweets (stratified across languages to ensure sufficient representation) and found, using Google's language auto-detection that only 34 were mislabeled, leading to an accuracy of 94.26%. Hence, to a first approximation, using Twitter's language meta-data is justified for the studies in the paper.

3.2. Descriptive statistics on language and prevalence

As a first step, we provide some descriptive usage or *prevalence* statistics for emojis in our dataset. Specifically, in Fig. 3(a), we plot (on the log–log scale) the number of tweets containing an emoji versus the number of emojis in the tweet. The goal is to show tweeting behavior with respect to emojis. Each emoji occurrence within the tweet is treated as unique for the purposes of Fig. 3(a) i.e. a ❤️ and a 😀 emoji within a tweet would lead to an emoji count of 2 for that tweet, as would two 😀 emojis. The figure clearly illustrates a power-law distribution, not unlike the degree distribution plots in social and information networks [39,40]. The vast majority of tweets only have one emoji associated with them (the point corresponding to 0 on the x-axis since $ln(1) = 0$).

Fig. 3(b) illustrates a slightly different perspective of emoji prevalence. The y-axis now shows the number of unique emojis that are used across all tweets for a given language.[5] For some languages, the total number of unique emojis is quite small, which may be due to data sparsity for that language. However, even for languages where many tens (or even hundreds) of thousands of tweets were observed (such as Spanish or 'es' on the x-axis), not all unique emojis available at the time were used. The decline is roughly linear on the rank-order plot, a surprisingly stable observation over the languages for which we have sufficient data (a minimum of 10,000 tweets with emojis were observed for each language in the corpus, and on the x-axis of Fig. 3(b)). This is the first indication that emoji usage may provide a unique window into language, national and cultural manifestations on Twitter, in line with

[5] Expansions for both language and country codes, which are used throughout this article, are provided in an Appendix at the end of the article.

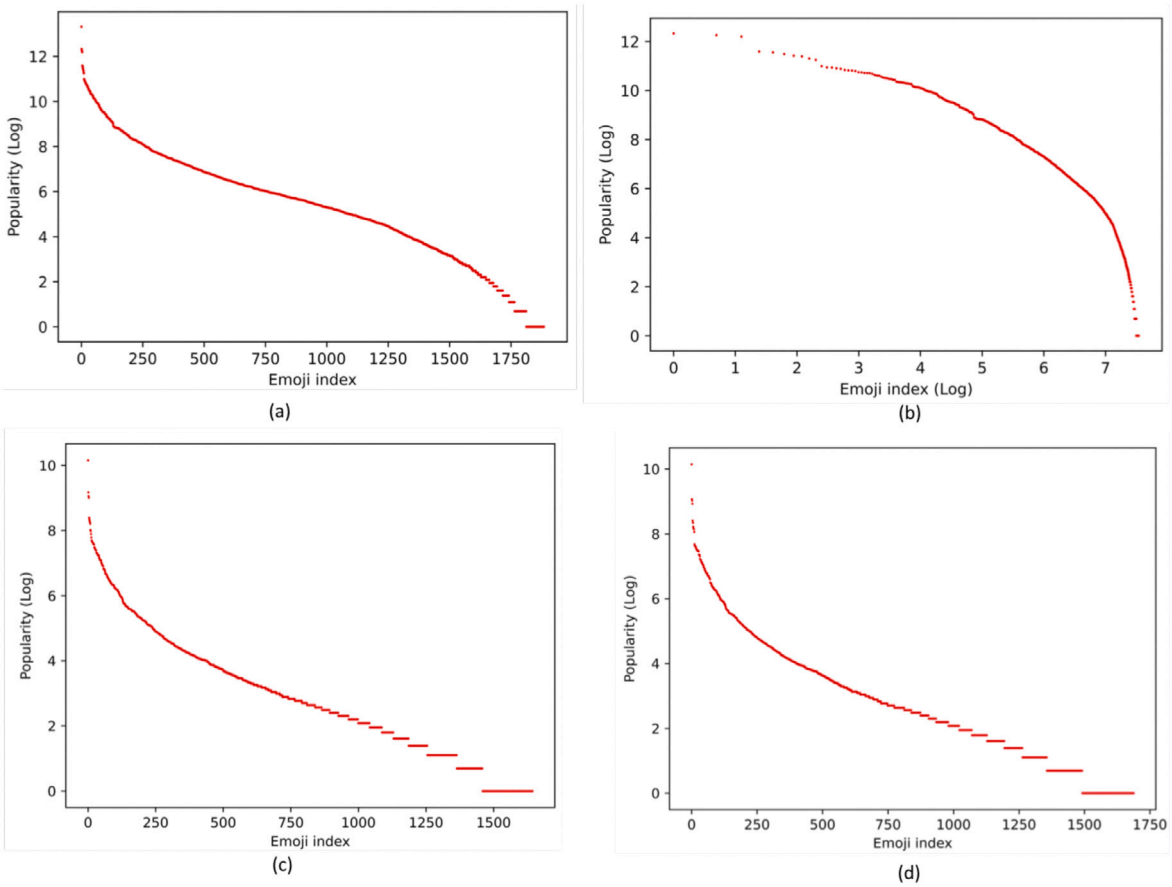


Fig. 5. A 'popularity' rank-distribution plot of emojis. (a) is a distribution of emoji popularity, which is the number of tweets (on the natural log scale) that contain the emoji referenced by an index, and with indices ordered from left to right in decreasing order of popularity (rank-ordered plot); (b) expresses the same data as (a) except that the x-axis is also on the log scale. Even on a log–log scale, the trend is clearly non-linear. (c) and (d) were constructed using the same methodology as (a) but only over a day's worth of data each (2 October, 2016, and 10 October, 2016, respectively, both of which were randomly selected from our corpus). Log equivalents of these plots look similar to (b), just as the trends in (c) and (d) closely mirror that of (a) and are hence not reproduced herein.

a prior study that only looked at culture [22]. The next few sections will explore this broad hypothesis in more precise detail.

Since there is a chance that the different numbers of tweets observed for different languages could cause a bias, we did an additional experiment whereby we randomly sampled 10,000 tweets for each of the top 30 languages, and computed similar prevalence statistics as shown in Fig. 3(b). As illustrated in Fig. 4, even after controlling for size, we continue to observe a difference across languages, although the relative rankings (the x-axis) of Figs. 3(b) and 4 are different. Therefore, the number of tweets does have an impact but the earlier suggestion that there are differences between languages still continues to be supported, even in the controlled setting. The experiment also suggests that using an information-theoretic measure like the KL-Divergence (which we describe subsequently) is an appropriate methodological choice for comparing emoji usage across languages, since by relying on probability distributions (rather than absolute values), the KL-Divergence is better able to better control for the dependence on sample size.

3.3. Results

3.3.1. Emoji popularity

The power-law distribution of emoji prevalence in tweets in Fig. 3 (a) suggests that the counts of emojis are skewed, with the majority of tweets containing very few emojis. However, there are also hints that not *all emojis are equal*. Intuitively, we observe this in everyday communication (at least on social media) by the heavy presence of the 😀 or other symbols (such as ❤️) compared to (say) the flag of a country. Our first research question attempts to study this issue systematically by asking, *how is language correlated to the popularity of an emoji, and how does it relate to a global distribution of emoji popularity?* We are especially interested, not only in determining how the 'popularity' of emojis compare, but if there is a consistent trend over short timescales and across languages.

Fig. 5(a) illustrates a 'rank order' plot where we assign each unique emoji an identifier (ID) in decreasing order of popularity (defined simply as the number of tweets, plot on the natural log scale, that contain at least one occurrence of the emoji assigned that ID), and then plot the popularity of the emoji against the emoji's ID. Fig. 5(b) shows the same data but the x-axis is also on the log scale. To further support the finding that the trend is neither an aberration nor an anomaly of a single dataset, we isolated two days' worth of data from our corpus by randomly selecting two days (October 2, and October 10) and plotting the data again for each of those two days (Figs. 5 (c) and (d)). While the data becomes sparser towards the higher IDs (or ranks), likely due to the absence of enough data over one day for the least popular emojis, the same trend as in Fig. 5(a) emerges fairly quickly. Furthermore, the fact that it does not change much from one day to twenty nine days attests to its general stability. One possibility is that, in the 'dialect' of social media, the most popular emojis are the equivalent of common words in ordinary language, the usage patterns and frequency of which emerge fairly quickly, along with their relation to one another.

Despite its stability, we note that neither the trend in Fig. 5(a) nor Fig. 5(b) corresponds to a 'standard' power-law or Heap's law distribution. Variations of the distribution do not yield these laws either. However, a large span of the function exhibits linearity and the steep non-linearities occur towards the *beginning* and the *end* (representing

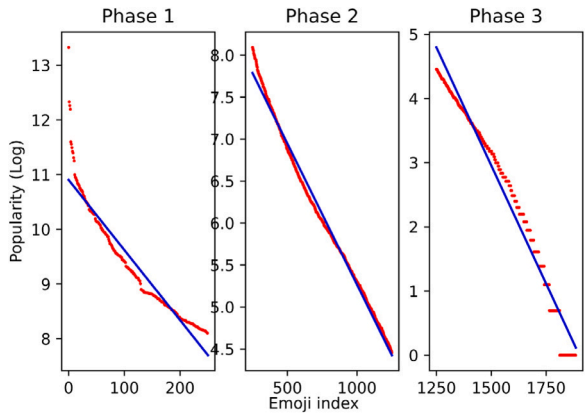


Fig. 6. A decomposition of Fig. 5(a) into three 'phases', where the middle phase (spanning the vast majority of emojis i.e. about 1000 of approximately 2000 emojis) best obeys a power-law distribution. The power-law coefficient (slope of solid line) in all cases is very low, ranging from −0.013 (the left plot) to −0.003 (the middle plot). The goodness of fit (R^2) in all cases is more than 90%. Note the differences across both the x and y-axes between all individual phases.

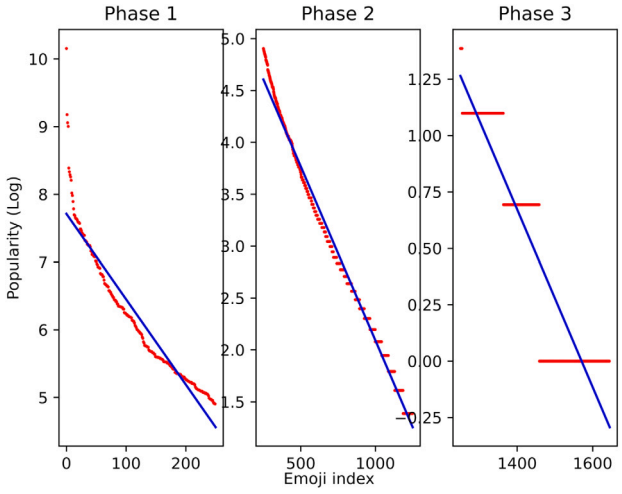


Fig. 7. A decomposition of Fig. 5(c) (Oct. 2) into three 'phases', where the middle phase best obeys a power-law distribution. The coefficients are remarkably stable compared to Fig. 6. Phase 3 is an exception likely due to extreme data sparsity in the low-popularity regime in that phase for the one-day data.

the *most* and *least* popular emojis, respectively). We surmise, therefore, that the curve can be 'decomposed' into three separate *phases* each of which can be made to approximate a power-law. To study the merits of this decomposition, we reproduce it in Fig. 6 (similarly, for the one-day distributions in Figs. 5(c) and (d) we plot the 3-phase distributions in Figs. 7 and 8).

In comparing the month-long and one-day distributions between Fig. 6, and Figs. 7 and 8 respectively, we note that there is a strong similarity between their first and second phases in all figures. The coefficients for Phase 2 across all plots are virtually identical, with a difference of 0.001. The choice of day does not matter much, as the similarity between Figs. 7 and 8 (for Oct. 2 and Oct. 10, respectively) shows.

Furthermore, as evidence of the persistence of this trend in the emoji data, we plot it separately for (tweets corresponding to) the six top languages in Fig. 9. The trends in Fig. 9 look virtually identical to the ones illustrated earlier in Fig. 5(a), (c) and (d), although details differ depending on the language and the extent of its prevalence in the data. We hypothesize that some emojis are both more universal in scope, and

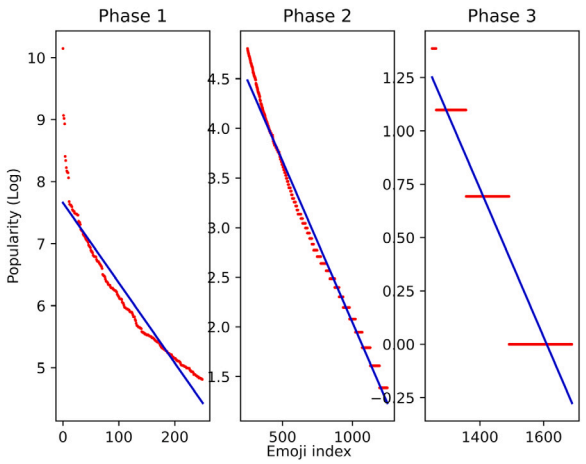


Fig. 8. A decomposition of Fig. 5(d) (Oct. 10) into three 'phases', where the middle phase best obeys a power-law distribution. The coefficients are remarkably stable compared to Fig. 6. Phase 3 is an exception likely due to extreme data sparsity in the low-popularity regime in that phase for the one-day data.

globally popular, in most countries and languages. Others may have country- and language-specific dependencies, as later discussed.

These empirical illustrations seem to be confirming the presence of a strong macroscopic phenomenon with respect to the popularity of emojis. The 3-phase structure, which demarcates the emojis in terms of 'highly popular', 'moderately popular' and 'low popularity' groups is one possible explanation. Even within those phases, the specific distributions or coefficients observed leave much to be explained by a more fundamental sociological theory of emoji meaning and use. Although we do not pursue such a study here, we studied the emojis that were falling within each of the three phases and found that there were some within-phase similarities that match everyday expectations. Among the 'highly popular' emojis were mostly smileys and other 'facial' emojis (for expressions such as anger and sadness). Emojis representing certain popular sports, such as basketball, soccer, baseball and rugby were also in that phase. In contrast, the second phase ('moderately popular') was found to be far more heterogeneous, while the third phase mostly had the flags of countries,[6] although it was also heterogeneous. While a full quantitative study of phase-specific trends is not in the scope of this study, we do provide some data in Fig. 25 in the Appendix, whereby we use an emoji's classification (e.g., 'activity' vs. 'smileys and emotion') to graphically categorize some of the differences between the phases. The key point to note, which is commensurate with Fig. 6, is that the three phases do exhibit significant differences. One hypothesis is that they play different roles in online communication, and also, that they are more (or less) accessible on mobile devices, requiring more deliberation on the part of the user who is tweeting the emoji. In the future, this phenomenon and hypothesis could be studied in greater detail.

3.3.2. Linguistic dependencies

Earlier, we had plot the number of distinct emojis per language in our dataset in Fig. 3(b). The plot illustrated that language does make a difference, and it would seem that some languages are more constrained in emoji usage than others. Data sparsity may be cited as one explanation, but it cannot explain all the variance among languages, since we retained the top 30 languages that each had at least 10,000 recorded tweets. This is also suggested by the additional results in Fig. 4. Furthermore, Fig. 9 in the previous section would seem to

[6] However, there were many flags that were also in the second phase. Only the US flag was in the first phase.

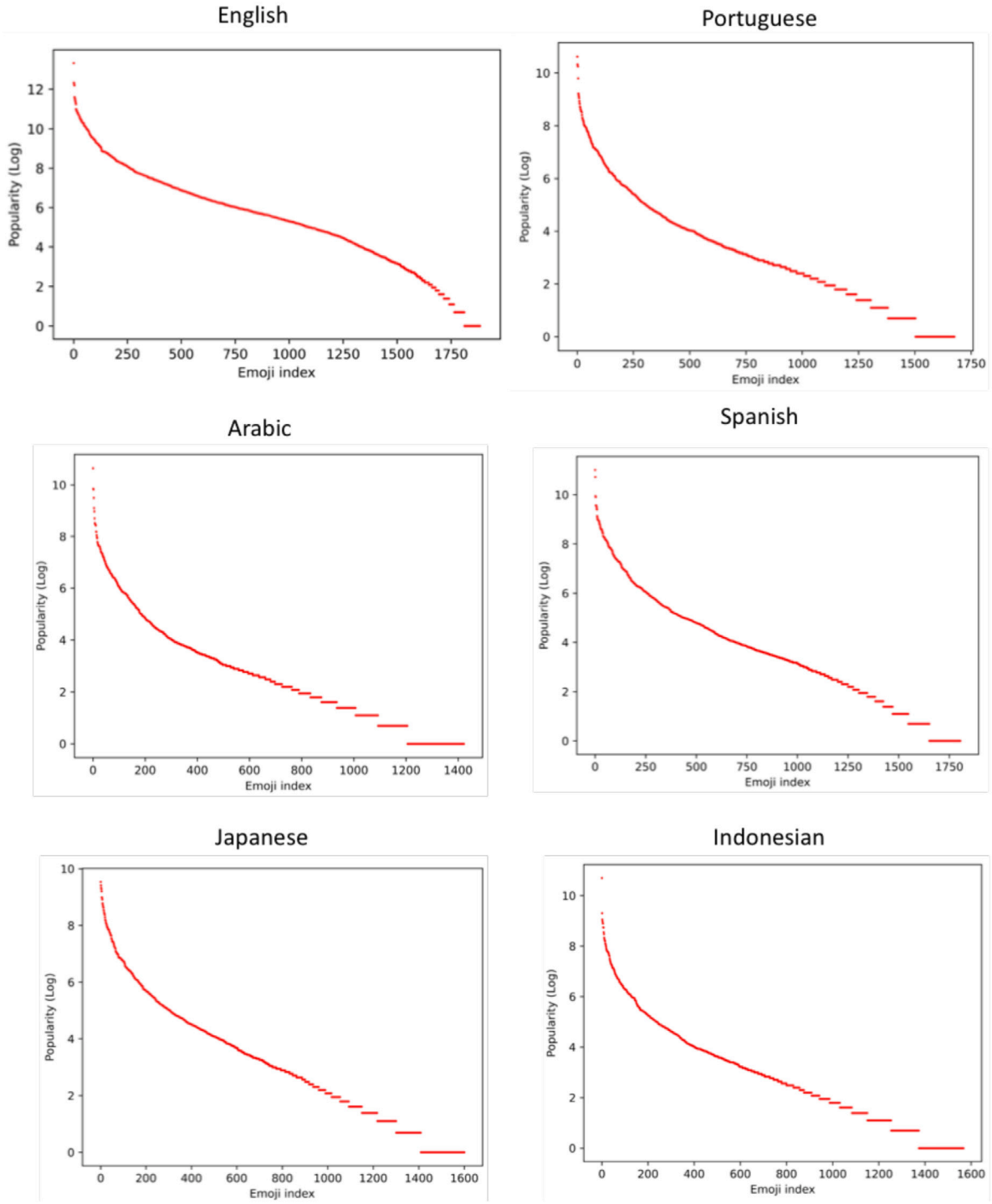


Fig. 9. A distribution of emoji 'popularity' over the month-long data, with popularity defined as the number of tweets (on the natural log scale) that contain the emoji referenced by an index, and with indices ordered from left to right in decreasing order of popularity (rank-ordered plot). Data is displayed for tweets in the top six languages (see Fig. 3(b)).

suggest that languages are more similar than different, and that some trends are relatively stable.

In this section, we study our second research question, namely, *to what extent, if any, a quantifiable dependency exists between language and emoji usage?* As a first step, to understand the importance of *unique* emojis (and relative usage of emojis) we contrast Fig. 3(b), which only considered count of unique emojis, by plotting in Fig. 10, a logarithm of *total* emoji counts. In considering the plot, it is fairly evident that the higher total number of emojis for tweets in languages such as English can be partially explained by the volume of tweets observed in that language, which is an expected finding. In fact, when we do the same experiment that we had done in Fig. 4 (i.e., computing the total numbers of emojis only for 10,000 tweet samples per language, while controlling also for the effects of randomness), we find in Fig. 11 that the differences are greatly dampened (although the usage of the log scale can also dampen differences). This suggests that, while emoji usage across languages is similar in certain ways (such as in the total number of emojis used, but also *distributionally*, as noted in the previous section), there are also differences. In this section, we attempt to better understand these differences.

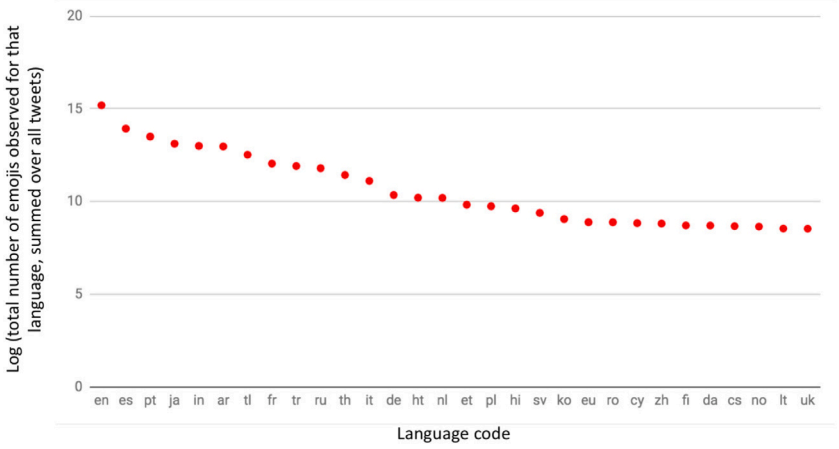


Fig. 10. A distribution of *total* emoji counts versus language, in decreasing order of y-values (rank-order plot).

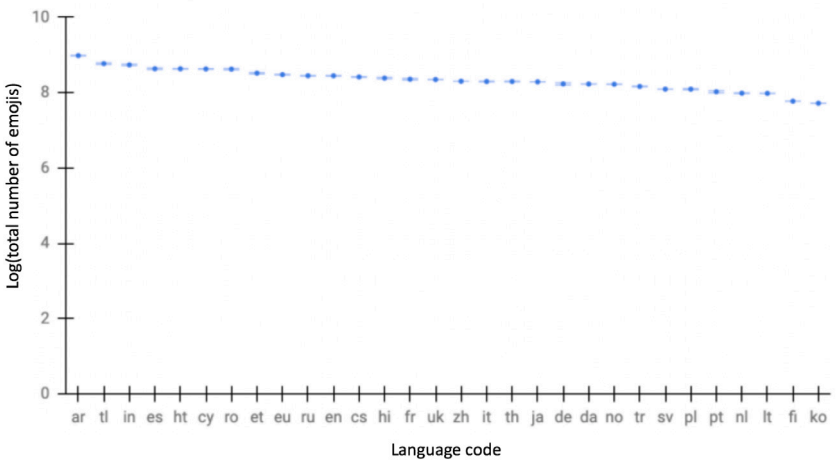


Fig. 11. A distribution of *total* emoji counts versus language, in decreasing order of y-values (rank-order plot), but using only 10,000 randomly sampled tweets per language (with 10 iterations to control for the effects of randomness, similar to the methodology used earlier for Fig. 4).

To get more fine-grained insights into a linguistic correlate of emoji usage, we compute the *Kullback–Leibler (KL) Divergence* between the usage probability distributions of every pair of languages. The KL-Divergence is a measure of how one probability distribution is different from another (reference) distribution [41]. It is an information-theoretic measure. The higher the KL-Divergence the more different the two distributions are. The KL-Divergence has a value of 0 if both distributions are identical.

Clearly, in order to use the KL-Divergence, we need to express the emoji usage as a *language-specific* probability distribution.[7] The specific methodology is as follows. Given a language l, we define the *language-specific usage* $U_l(e)$ of a *particular* emoji e as the *sum of weighted occurrences* of that emoji, divided by the total number of tweets in that language in our corpus. By weighted occurrence of an emoji in a tweet, we mean the following: given an emoji e, the weighted occurrence of e with respect to a tweet t is 1.0 if it is the only unique emoji in that tweet, and 0 if it does not occur in the tweet at all. However, if there is more than one unique emoji in the tweet, the weight is the reciprocal of the number of unique emojis in the tweet. For example, if there are three unique emojis in the tweet, then each of the emojis gets a weight of 0.33. By adding all of these weights (per emoji) across all tweets (whether in the context of a language or country), and dividing by the total number of tweets (with that language or country), we get a *usage probability distribution* over all emojis for each language and country.

Given the usage probabilities calculated for every language as described above, the question arises as to how different the distributions of emoji usage are with respect to pairs of languages. If all languages use emojis in very similar ways, the intuition is that the 'difference' between the usage probability distributions would be small. As discussed above, a well-defined way of computing such differences between two probability distributions is the KL-Divergence measure. Given two probability distributions $P(x)$ and $Q(x)$ with the same support, the KL-Divergence is formally defined as:

$$D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) log(\frac{P(x)}{Q(x)}) \quad (1)$$

In all cases where we use the KL-Divergence, the 'support' (or the x-axis of the probability distribution[8]) is always the set of unique emojis. Specifically, given a language l_1 and a language l_2 (a similar analysis applies in subsequent sections for *countries*), we proceed in the following way. First, we filter the support to be the set of emojis that have non-zero occurrences in both l_1 and l_2; furthermore, to avoid issues due to data skew, we only consider emojis that have a minimum usage probability of 0.003 in each of the two languages. This leads to discarding of emojis (for that language pair) where no (or few)

[7] A similar methodology applies for country.

[8] Equivalently, the 'event space' over which probabilities are computed.

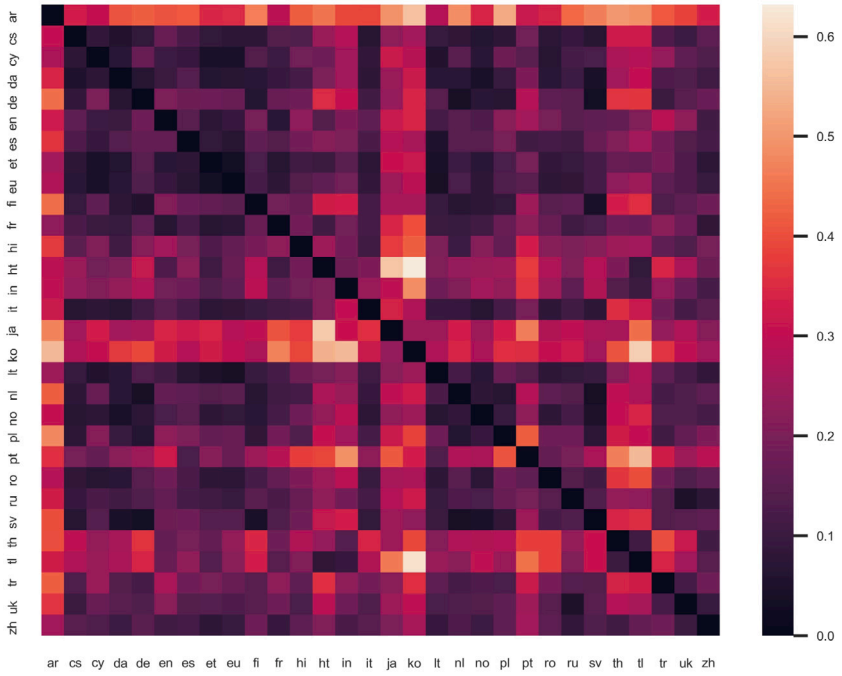


Fig. 12. A 'heatmap' expressing the KL-divergence between the usage distributions of two languages. Per the definition of KL-divergence in the text, the row is consistently treated as the reference distribution ($P(x)$). The map is asymmetric since the KL-divergence is asymmetric. Rows and columns should not be directly compared, but in individually comparing the sets of rows together and the sets of columns together, similar conclusions can be drawn from both comparisons. Note that a higher KL-divergence indicates (intuitively) more *dissimilarity* between the usage distributions of the two languages representing the cell.

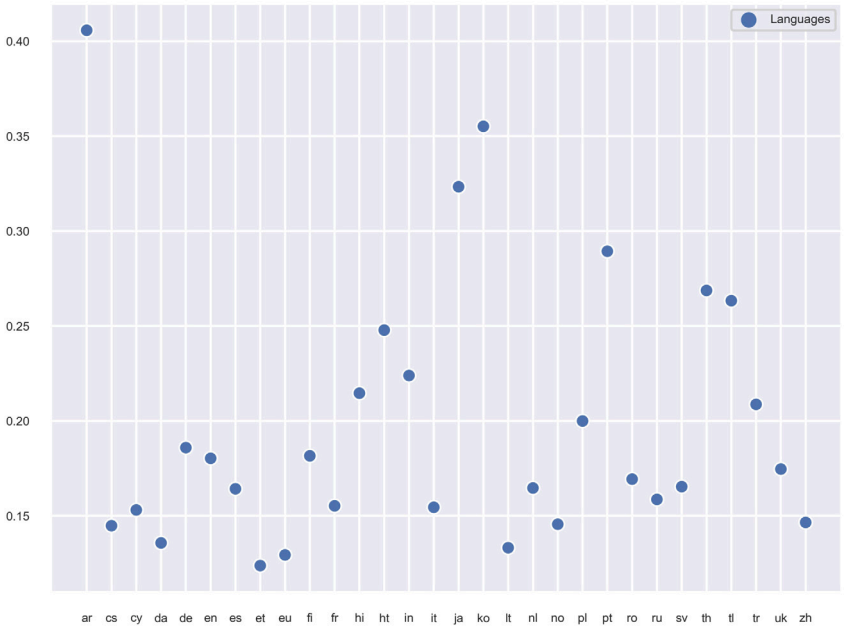


Fig. 13. The *mean* KL-divergence of the usage popularity distributions of all other languages compared to the distribution of the language (on the x-axis). A high value (such as for Arabic) indicates that the language is divergent from most other languages in terms of emoji usage. Low values (such as for Danish and Basque) indicate the opposite.

values were observed for either language and were found to lead to stable distributions and KL-Divergence measurements. By applying this filter, we were able to limit the support to the top 3% emojis (since, on average, for a given pair of languages, l_1 and l_2, about 20–60 emojis met the constraint that the usage probability not only had to be non-zero, but greater than 0.003 for both languages). The filter addresses a methodological concern that may arise that differences in KL-Divergence may get amplified potentially due to data sparsity or skew (which affects the probability distribution), even though the KL-Divergence itself is not sensitive to absolute values or scales as it only takes probability distributions as inputs.

The KL-Divergence between two distributions is a single number between 0 and 1, with a higher value indicating greater divergence (intuitively, dissimilarity in usage). We plot the pairwise KL-Divergences between the distributions using a heat map (Fig. 12). Note that the

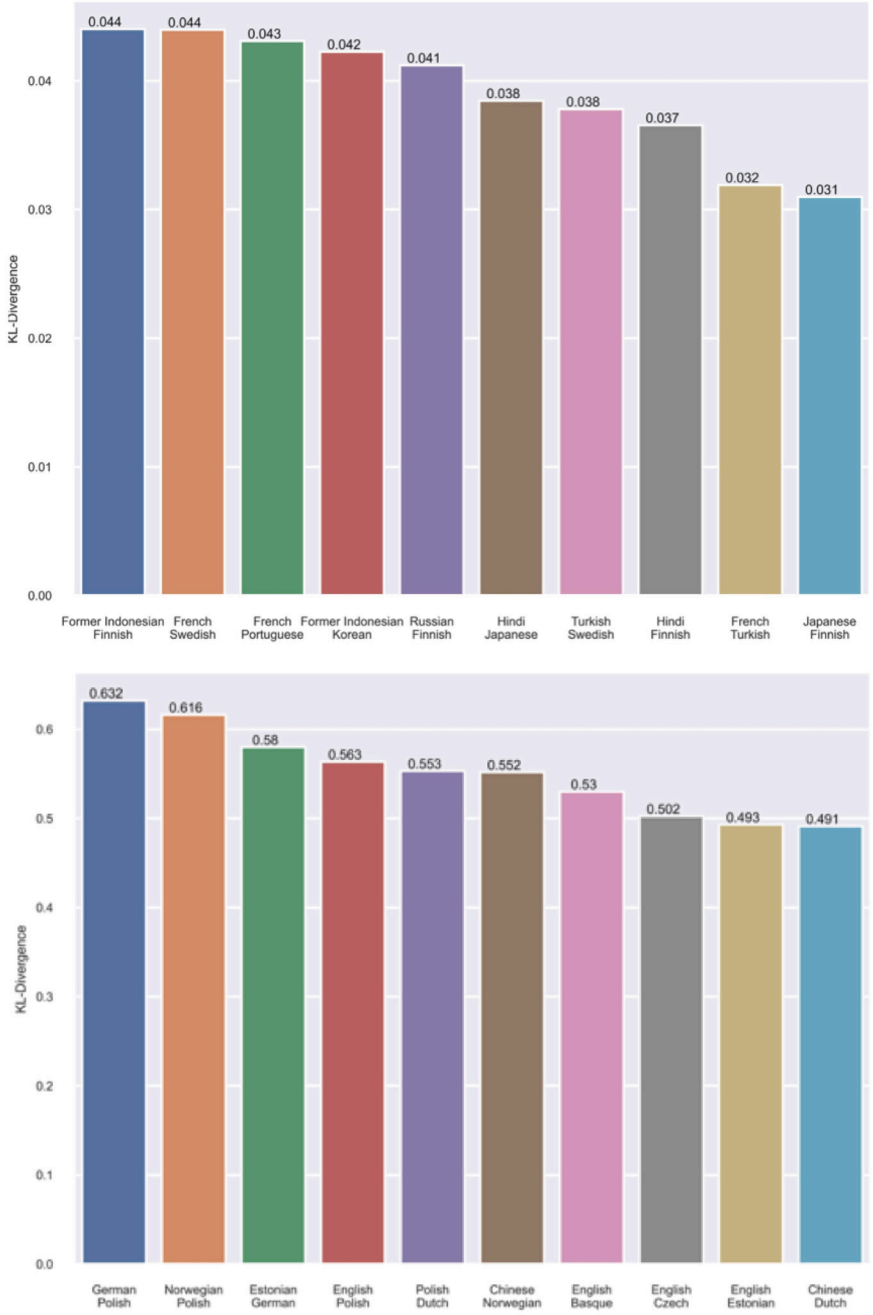


Fig. 14. Some selective results from Fig. 12 of pairs of languages that are very similar to (top) or different from (bottom) each other in terms of emoji usage. Note that *y*-axis scales are different between the two figures to magnify relative proportions among language pairs in the two figures.

matrix is asymmetric since the KL-Divergence changes once the two distributions $P(x)$ and $Q(x)$ are reversed in the equation above. However, in studying the figure, the overall conclusions and relative comparisons do not change. Much of the matrix is 'dark', implying that generally speaking, divergence is low and languages are relatively similar in terms of their emoji usage. However, we note that there are 'pockets' of high-divergence cells and rows in the matrix, which shows that this is not a universal social law, and some languages are very different in emoji usage than others. One possible explanation is an underlying cultural or geographical variable that could explain some of this variance. More work is needed in the future, however, to tease apart such effects.

Fig. 13 allows us to study the 'average' divergence of a language's usage probability distribution to that of all *other* languages. This is easily computed by taking, for each language, the mean of all KL-Divergences in the row corresponding to that language in Fig. 12 (except the diagonal element, which corresponds to the language itself). Fig. 13 shows that there are indeed outliers in this distribution. In particular, Arabic has the most different (in terms of the adopted mean KL-Divergence metric) usage probability distribution, compared to other languages, among the languages in our set, followed by Korean (ko), Japanese (ja), Portuguese (pt), Thai (th) and Tagalog (tl). There is higher clustering of mean KL-Divergences for other languages, although some still show much higher mean KL-Divergence than others. Assuming that these are not transient effects, but are replicated in similar studies conducted in other time periods, we suspect that a geographical or cultural variable may again help explain some of the variance.

Delving further into the findings of Figs. 12 and 13, it is also instructive to compare the most similar and dissimilar *pairs* of languages in terms of their emoji usage probability distributions. The top figure in Fig. 14, where the *y*-axis (the KL-Divergence between

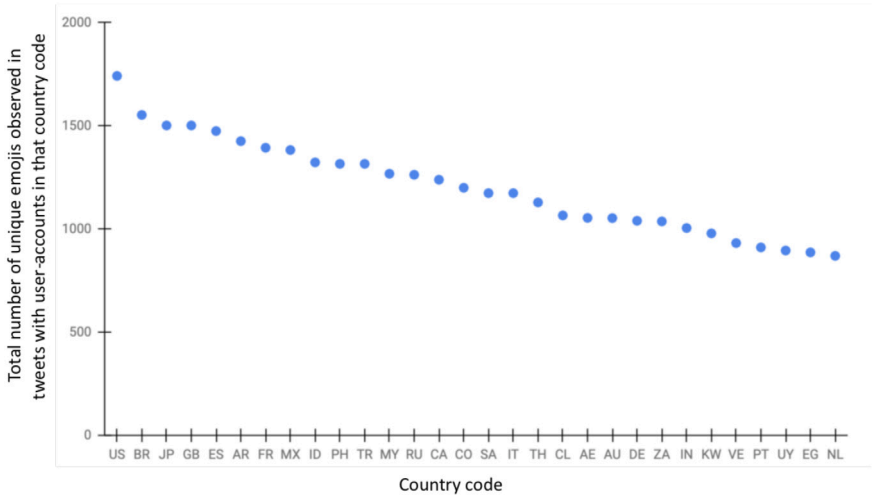


Fig. 15. A rank-order plot of the number of unique emojis recorded in the corpus per country. As with the methodology applied to the language-dependency analysis, we retain the top 30 countries to avoid data sparsity. We have verified that the thirtieth-ranked country in the corpus has at least 10,000 tweets associated with it.

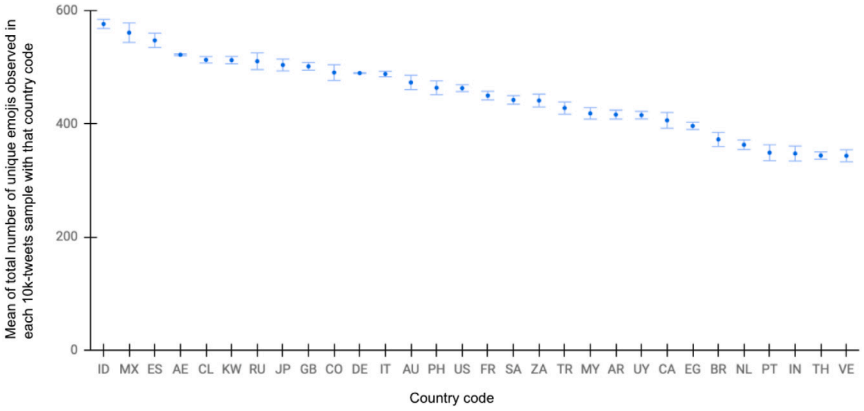


Fig. 16. The number of distinct emojis recorded in the corpus per country, for the top 30 countries in our corpus (rank-order plot). Unlike Fig. 15, we controlled for different numbers of tweets per country by randomly sampling 10,000 tweets per country. To control for the effects of randomness, we conduct 10 iterations, and report with error bars.

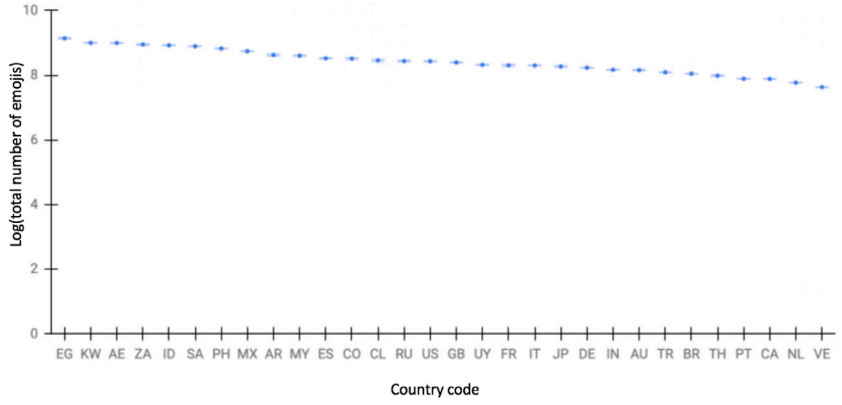


Fig. 17. The logarithm of the total number of emojis recorded in the corpus per country, for the top 30 countries in our corpus (rank-order plot). We controlled for different numbers of tweets per country by randomly sampling 10,000 tweets per country. To control for the effects of randomness, we conduct 10 iterations, and report with error bars.

the pairs of languages shown for that bar on the x-axis) only ranges from 0-0.044 (indicating low divergence and high similarity between usage probability distributions), indicates that the most similar,[9] pair of languages is Japanese and Finnish, followed by French and Turkish. These are interesting findings because the languages are *linguistically* dissimilar. In contrast, looking at the bottom figure in Fig. 14 we find that languages such as German and Polish are quite divergent overall, as are English and Czech. An interesting avenue for future research is to determine the variables (e.g., weather seasons) that are mediated through a language variable that could start to explain

[9] This pair occurs far on the right, and not left, of the top figure in Fig. 14 since the *y*-axis is still increasing in KL-Divergence.

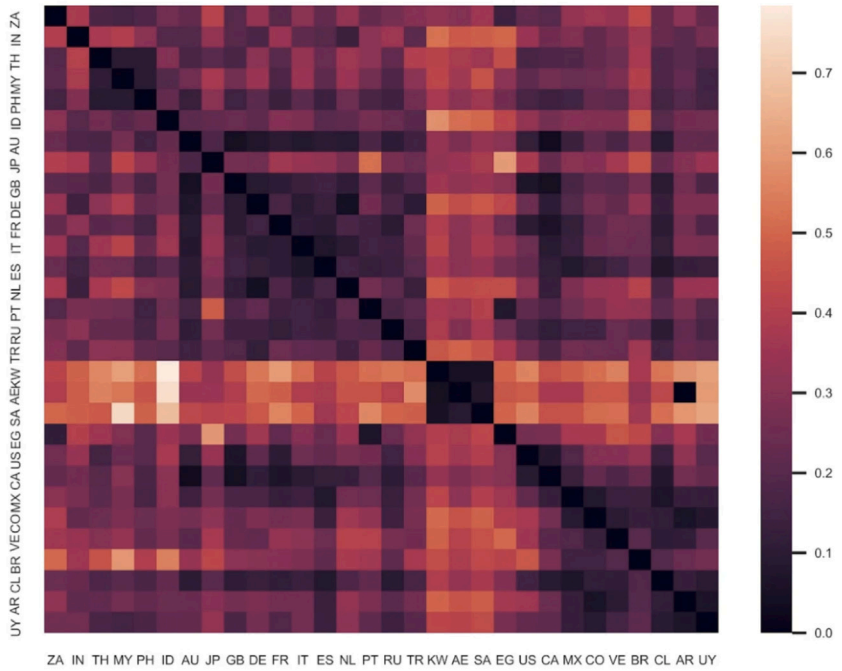


Fig. 18. A 'heatmap' expressing the KL-divergence between the usage distributions of two countries. The same methodology applies as noted earlier in the caption of Fig. 12. Countries are grouped by geopolitical region i.e., ZA is in Africa, IN, TH, MY, PH, ID, AU and JP are in the Asia-Pacific region, GB, DE, FR, IT, ES, NL, PT, RU and TR are in the European region, KW, AE, SA and EG are in the Middle East, US and CA are grouped together, and MX, CO, VE, BR, CL, AR and UY are grouped as Central and Latin American countries.

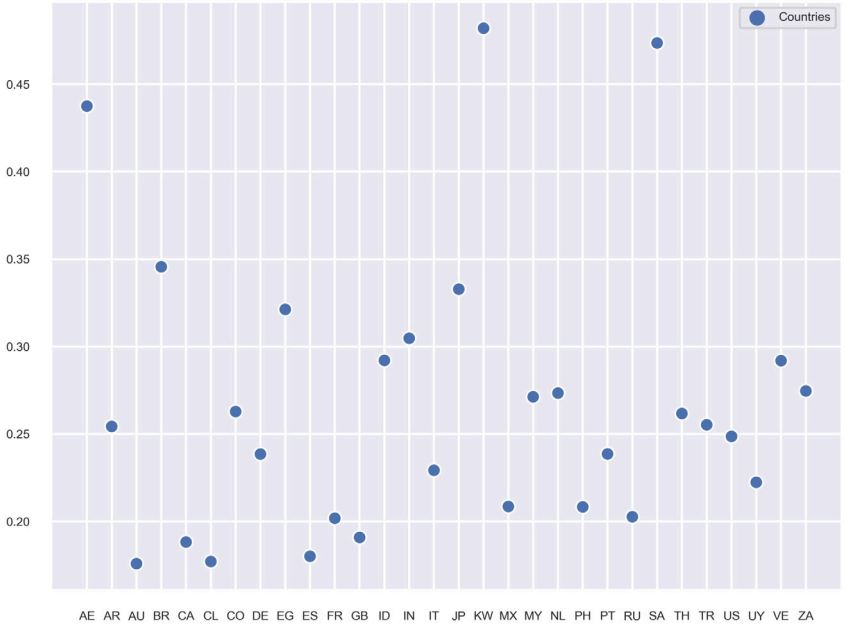


Fig. 19. The *mean* KL-divergence of the usage popularity distributions of all other countries compared to the distribution of the country (on the x-axis). A high value (such as for Venezuela and South Africa) indicates that the country is divergent from most other countries in terms of emoji usage. Low values (such as for Spain and France) indicate the opposite.

these usage similarities and dissimilarities. The size of the population tweeting in that language, for example, may (partially) explain some of the findings, but more controlled studies are required to investigate hypotheses with such high specificity in detail.

3.3.3. Country dependencies

The previous section discussed some of the empirical relationships between language and emoji usage. This section attempts to do the same, but with country instead of language. Specifically, we ask the research question, *to what extent, if any, a quantifiable dependency exists between country and emoji usage?* The core methodology is largely similar, permitting us to also draw some conclusions about whether language or country is the stronger determinant of emoji usage patterns.

As a first step, we plot the number of distinct or unique emojis observed per country in Fig. 15, for the top 30 countries, in a similar vein as Fig. 3(b) earlier. Once again, we find that a location feature (as well as language, from earlier results) serves as a kind of independent

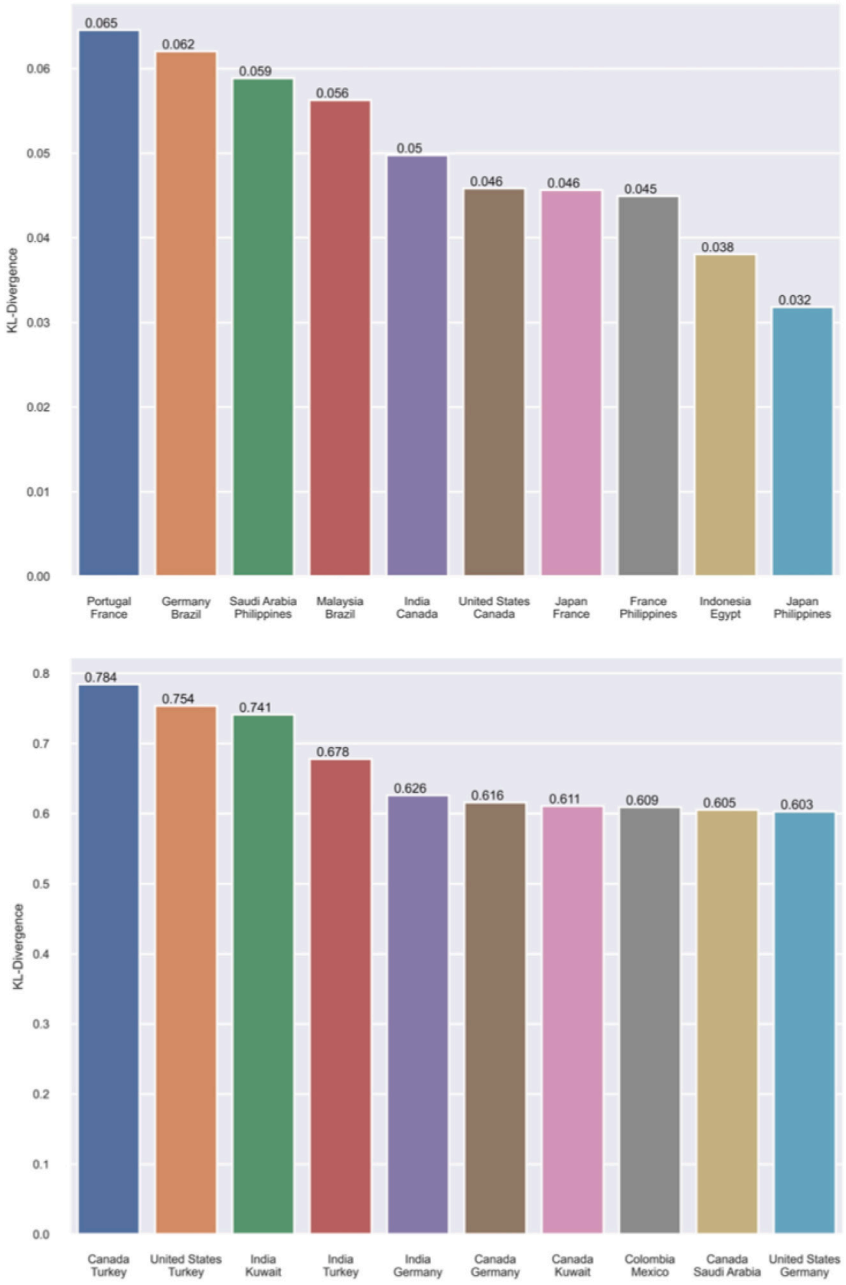


Fig. 20. Some selective results from Fig. 18 of pairs of countries that are very similar to (top) or different from (bottom) each other in terms of emoji usage. Note that *y*-axis scales are different between the two figures to magnify relative proportions among country pairs in the two figures.

variable when determining diversity of emoji usage in aggregate. A slight linear decline is again observed, similar to Fig. 3(b). We also conducted the experiment with 10,000 tweets (randomly sampled) per country, and find that differences are observed even when controlling for size (Fig. 16). Similarly, we also computed a plot analogous to Fig. 11 (involving the log of the total number of emojis rather than the number of unique emojis, while controlling for size), and found the conclusions to be qualitatively similar (Fig. 17). Namely, the distribution of unique emojis is more distinctive across countries than the distribution of total emojis, which can (to a great extent) be explained by size. An alternative explanation is that, due to proliferation of languages such as English as modern-day 'lingua franca' in many countries, as well as the nature of online dialect itself, a person's country (rather than language) contributes more strongly to their online identity and allows us to observe stronger clustering effects.

Similar to Fig. 12, we plot a country-by-country KL-Divergence 'heatmap' in Fig. 18. Once again, we find that the majority of the heatmap is on the darker side, indicating low, rather than high, divergence (on average, for the majority of countries). However, we see that there are also some light-colored rows and columns, particularly for Saudi Arabia (SA), Kuwait (KW) and several other countries, as described below. Note also that, in the heatmap, we have grouped countries by geopolitical region (Middle East, North America, and so on). When viewed by region, we find that the Middle East has low divergence among the countries in that region (the 'black square' in the center right of the heatmap) but high divergence with many other countries, with few exceptions. To a lesser extent, this is also found with some of the countries in the Asia-Pacific region (Thailand, Myanmar and Philippines). Other results, discussed below, bear this finding out as well.

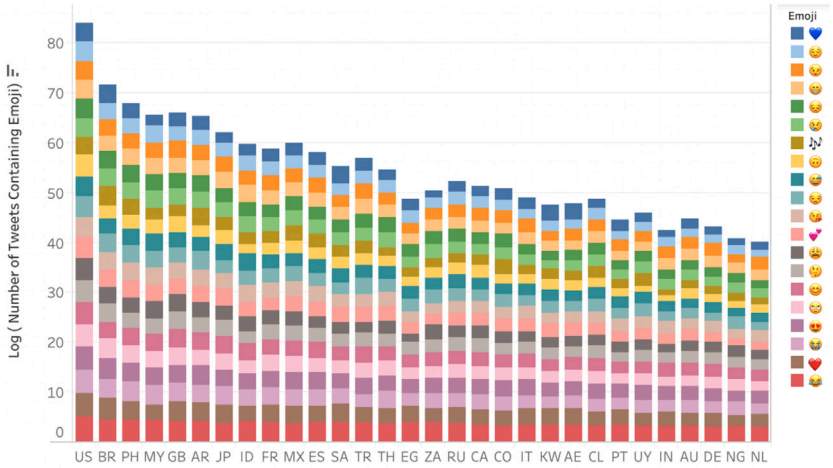


Fig. 21. A distributional breakdown of the most frequent emojis by country (with the *y*-axis on the log scale). The *y*-axis expresses the count of tweets per emoji per country that contain only that emoji (one or more times). We only consider the top 20 emojis, with the most frequent 'laughing till tears are coming out' emoji at the bottom of the legend. Other details are described in the text.

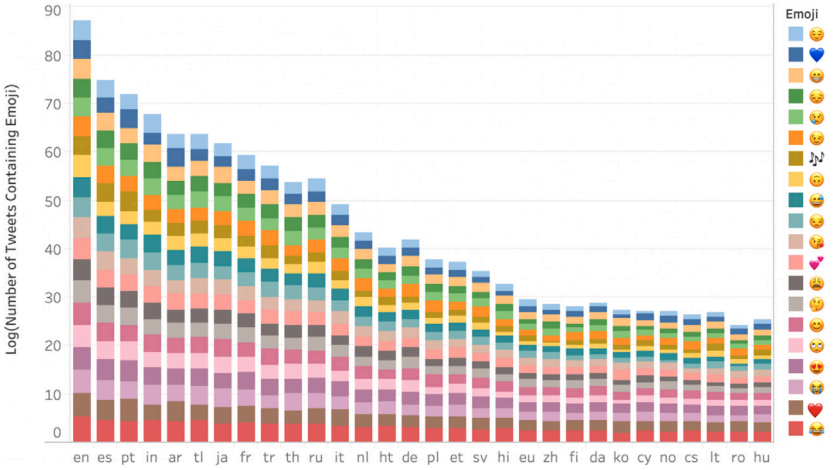


Fig. 22. A distributional breakdown of the most frequent emojis by language (with the *y*-axis on the log scale). We only consider the top 20 emojis. A similar methodology as Fig. 21 applies.

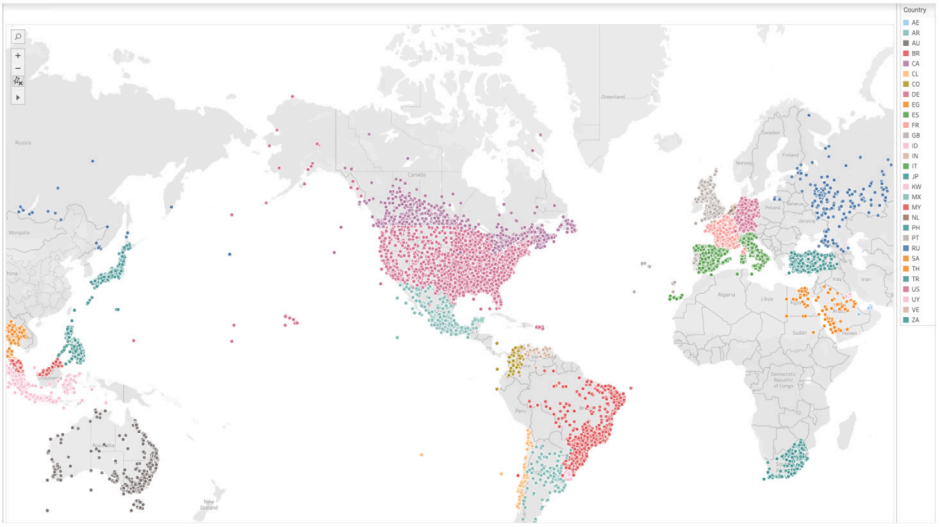


Fig. 23. A map of tweets in English that contain at least one emoji. We use geotags in the tweets for plotting the points. Recall that the corpus is limited to tweets in the top 30 countries (Section 3.3.3). All 30 countries (represented by 30 colors) in the corpus had at least some tweets in English. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

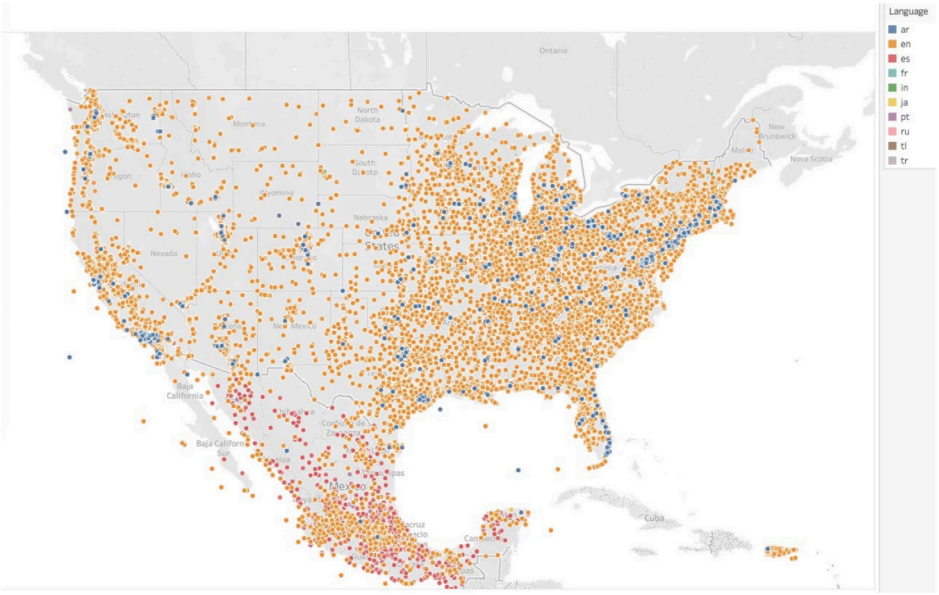


Fig. 24. A localized map of tweets in the US and Mexico. We use geotags in the tweets for plotting the points. We only show the top 10 languages in this region, as the data becomes too sparse after that (for these two countries).

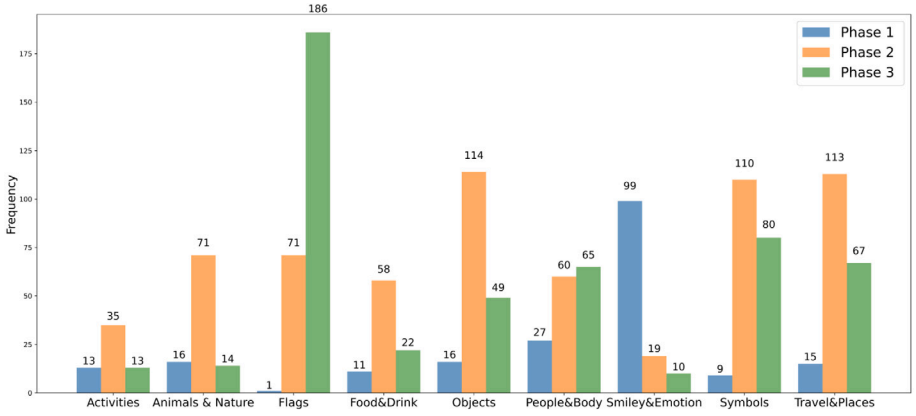


Fig. 25. Distribution of emojis by category for each of the three phases identified in Fig. 6.

As with Fig. 13 earlier, Fig. 19 allows us to study the 'average' divergence of a country's usage probability to all other countries. The methodology is identical to that adopted in the previous section. Recall that this computation was done by taking, for each country, the mean of all KL-Divergences in the row corresponding to that country in Fig. 18 (except the diagonal element, which corresponds to the country itself). Analogous to the findings in Fig. 13 earlier, Fig. 19 also indicates certain outliers. These include Venezuela (VE), Saudi Arabia (SA), United Arab Emirates (AE) and Kuwait (KW). While not strong outliers, Japan (JA), Brazil (BR), Venezuela (VE) also have high average KL-Divergence to other countries. European Union countries such as Spain (ES) and France (FR) have low average divergence with other countries. Are these patterns coincidences or are there fundamental features in these countries that might explain such patterns? For example, it may be that the lower divergences are observed because of heavy tourist traffic to these countries, rather than due to intrinsically low divergence of the local population. By comparing data between periods of low and high tourist activity (such as winter and summer), it may be possible for future work to tease apart the components of the divergence due to the local versus the visiting populace.

A comparison between Figs. 19 and 13 also reveals some striking details and confirms the findings earlier that we derived from comparing Figs. 18 and 12. In comparing Figs. 19 and 13, we have already seen that there are more outliers,[10] in Fig. 19 than in Fig. 13. This finding is also borne out by Fig. 18, which has 'lighter-colored' cells in greater numbers than observed in Fig. 13. Therefore, country, rather than language, contributes more variance to emoji usage globally. We hypothesized earlier that due to proliferation of modern-day lingua franca, such as English (especially in the commonwealth nations) and Spanish, a person's country and nationality may serve as a more distinctive marker of their social identity than their language. We leave it to future work to design specific studies to validate or refute such hypotheses.

Analogous to Fig. 14, we compare the most similar and dissimilar pairs of countries in terms of their emoji usage probability distributions in Fig. 20. There is some evidence of regionalism in the upper figure. For example, the United States and Canada, and Portugal and France have low divergence. Geographical distance seems to only weakly explain high divergence e.g., Colombia and Mexico have high divergence. While framing sociological hypotheses that may explain these findings

[10] It is instructive also in comparing the range of y-axis values in Figs. 13 and 19 to see this. In Fig. 13 the highest value is slightly above 0.4 and only two other points are above 0.3. In Fig. 19, seven points are above 0.3, and three are above 0.45.

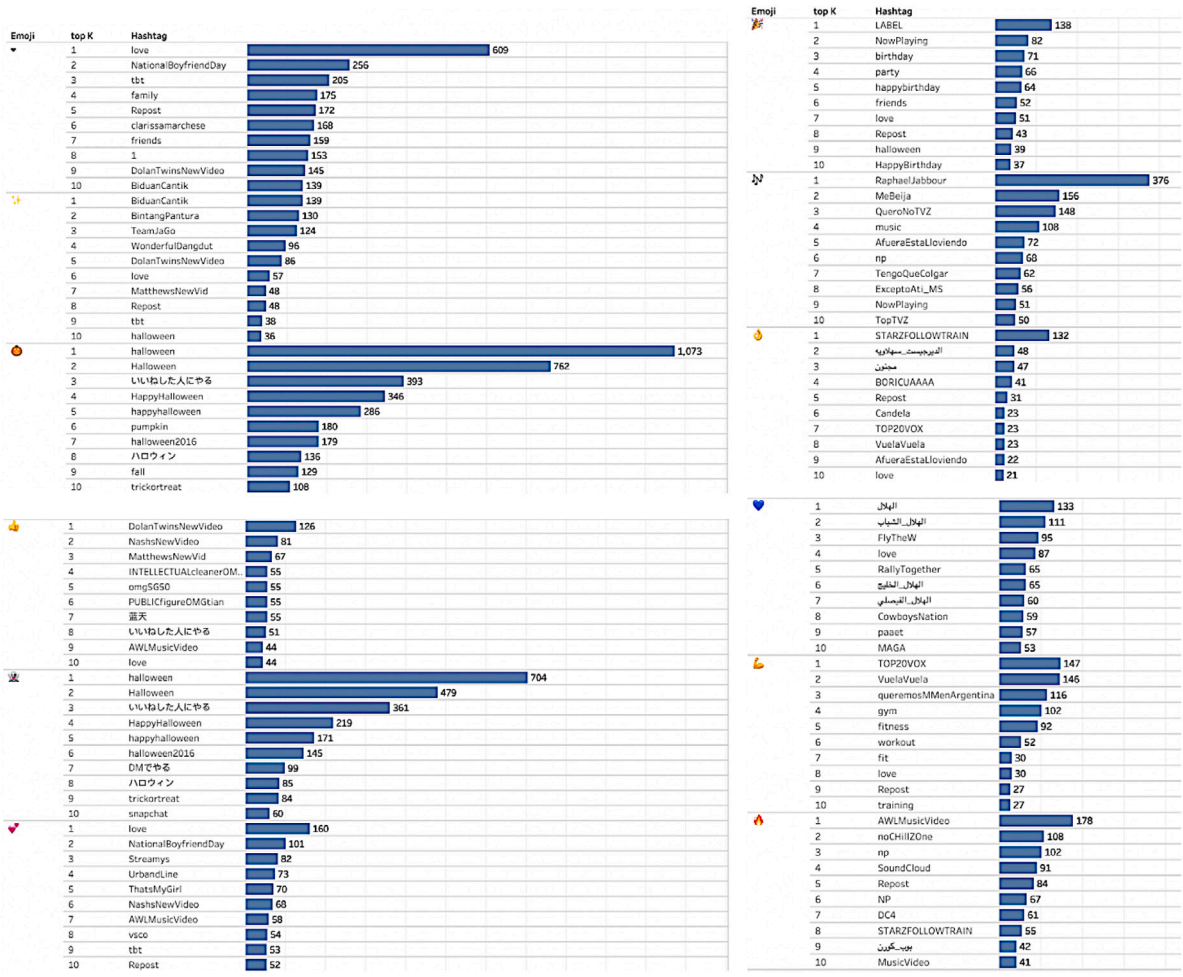


Fig. 26. Top-10 hashtag counts for a sample of popular emojis.

is beyond the scope of this present work, the findings here show that such a theory would have to be rich enough to explain when and why linguistic and geographic proximity of different populations matter (or does not matter). A candidate theory might, for example, rely on regression-based methodologies to quantify the effects of language, country, geography, politics, demographics and relevant cultural variables on a cross-sectional sample of tweets (with emojis) across the world.

4. Summary of experimental findings

Based on the studies and experiments in Sections Section 3.3, we summarize the following claims that seem to have the most robust empirical support:

1. Emoji *prevalence* (the extent to which emojis are used within tweets) follows a power-law distribution globally, but not all languages use all emojis (Fig. 3). Even after controlling for size, there are differences between languages (and also countries) in the numbers of unique emojis used.
2. Emoji *popularity* does not follow a single power law (or variant thereof, such as Heap's law) distribution but instead seems to exhibit a 3-phase structure (Fig. 5(a)). The vast majority of emojis fall within the second phase, which follows a power law distribution closely with coefficient 0.003 (Fig. 6). Not only is the trend robust to change of language (Fig. 9), but it emerges quickly even when using just one day's worth of data (Figs. 5(c) and 7).
3. The findings and visualizations in Sections 3.3.2 and 3.3.3 show that information-theoretic measures such as the Kullback–Leibler (KL) Divergence can help us to understand which countries and languages are divergent from which others, and also which ones are divergent on average from all others. Geographic, regional and linguistic differences may all serve as potential explanations for why we observe these differences, as may socio-economic and cultural variables. Some evidence of the latter was found in [22]. A future study may want to tease apart these influences using externally collected datasets.
4. Emoji usage has strong linguistic and country dependencies, but the latter has a stronger and more pronounced effect on emoji usage, as evidenced in comparing Figs. 18 and 12, and also Figs. 19 and 13. Languages that show high KL-Divergence compared to other languages, on average, include Arabic, Portuguese, Korean, Japanese, Thai and Tagalog. Western languages such as English, Spanish, German, French and Italian have KL-Divergences that are low, but not non-trivially so (between 0.15 to 0.2). Similarly, countries that are divergent from other countries in emoji usage include Venezuela, Saudi Arabia and Kuwait, while countries that generally have low divergence include Spain, the UK and France. The divergence of these countries is significantly lower (by almost 0.05) than countries such as the USA.

It is worthwhile comparing these findings briefly to two sets of findings previously cited in Section 2. The work by [22] found some evidence of cultural dependency, which they obtained using Hofstede's model of cultural dimensions [23]. However, our results suggest that a

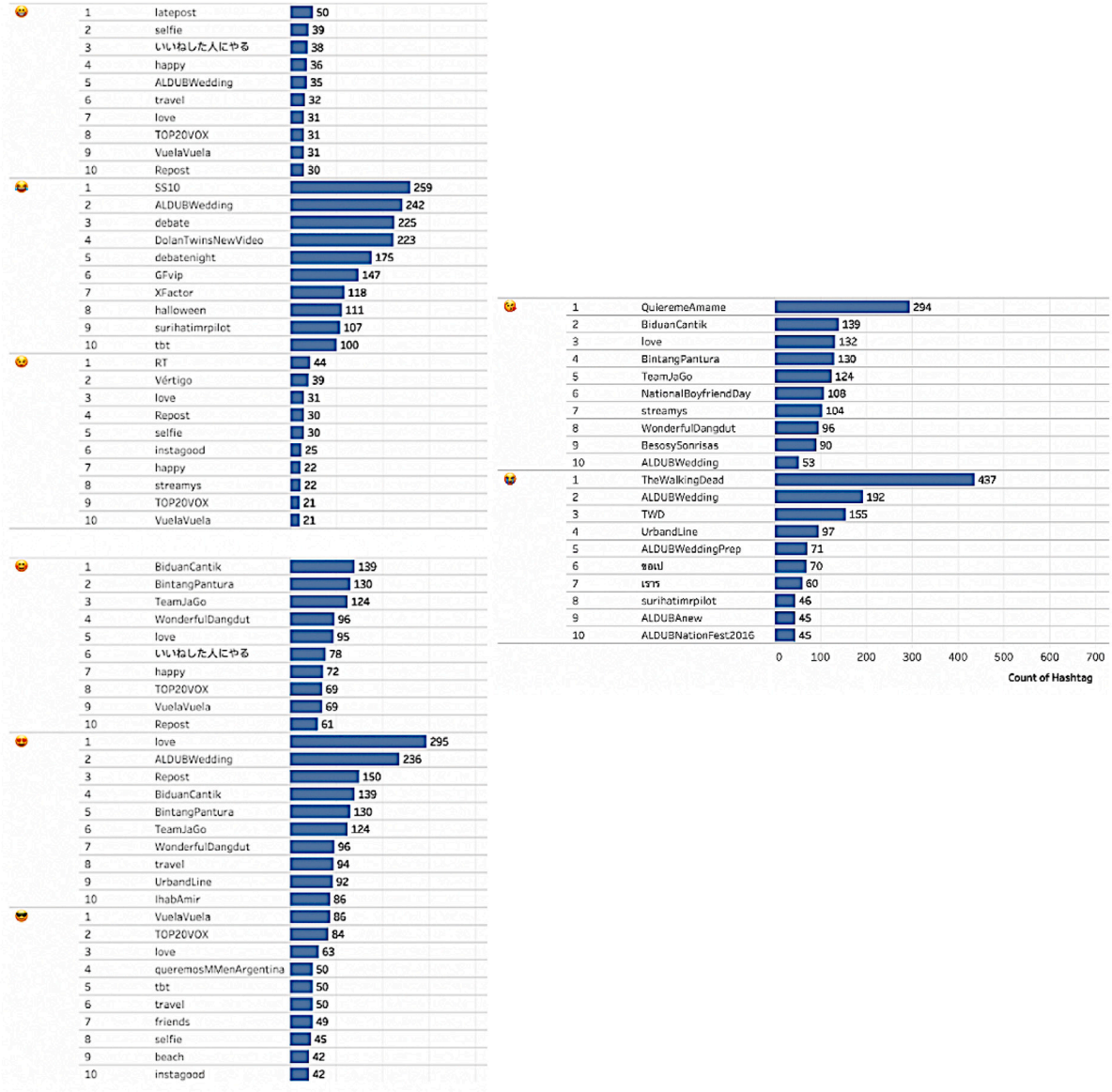


Fig. 26. (*continued*).

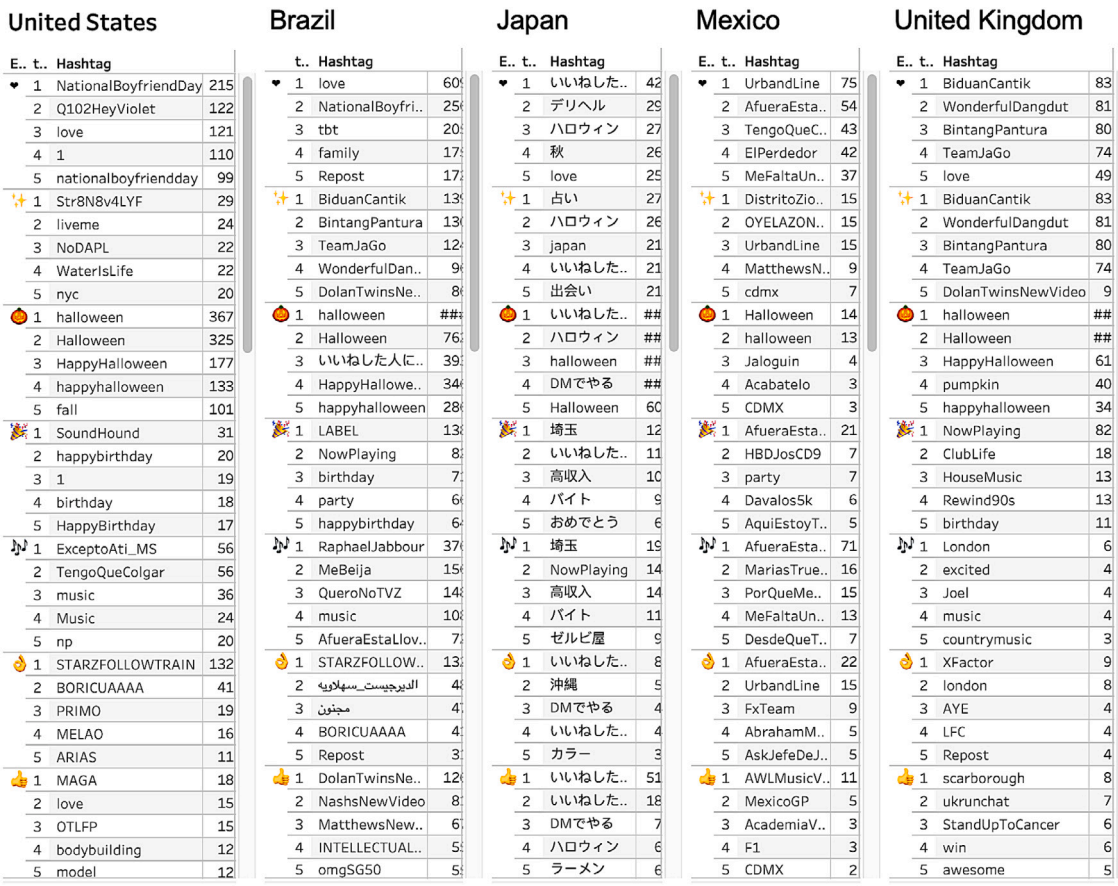
potentially different set of results may be obtained when language and country (both of which are themselves not necessarily independent of 'culture') effects are accounted for. The correlation between language, country and culture add support both to the findings in this paper and in [22]; namely, that there is an association between these variables and emoji use.

Similarly, [20] claimed to have detected correlations between a country's emoji usage and whether that country was a 'first world', 'second world' (and so on) nation. Our results show that this relationship does not hold once we disaggregate results and plot them at the level of individual countries (Fig. 19). In fact, the Western European nations (at least on the continent) seem to exhibit low divergence to one another and to other countries, while the USA has moderately-high divergence. Middle-eastern nations such as Saudi Arabia also have high divergence. When looking at nations that are developing, we see even more diversity: Mexico has low divergence, while both India and Venezuela have moderately high divergence. These results suggest that the manner in which [20] explains emoji usage (by aggregating countries in the way that the authors proposed i.e. by 'first-world' nations, and so on) could be supplemented by considering individual languages and countries.

5. Discussion

The previous sections showed the different ways in which language and country variables can influence the emoji usage distribution, as well as distributions and empirical laws concerning emoji distribution and popularity more globally. In this section, we supplement those analyzes with more qualitative observations. In studying Figs. 3 (b) and 15, it is evident that some emojis are favored (or put another way, other emojis are 'disfavored') in the context of some languages and countries than others. What is not evident is how the 'composition' of the emoji-set observed across languages compares to one another. For instance, are there universally popular emojis (such as the smiley face)? Are some emojis popular in some countries, but rarely used in others?

While detailed quantitative studies investigating these questions can also be designed, we take a more qualitative approach here. Fig. 21 shows a distribution of the most frequent emojis by country. To obtain the plot, we first isolated all tweets that contain exactly one unique emoji (though that emoji could be contained multiple times within the tweet), and then, for the top 20 most frequent emojis in the corpus, plot a per-country distribution. The results show that, while the 😂 emoji (which is at the end of the legend) is a popular one,



United States

E..	t..	Hashtag	
❤	1	NationalBoyfriendDay	215
	2	Q102HeyViolet	122
	3	love	121
	4	1	110
	5	nationalboyfriendday	99
✨	1	Str8N8v4LYF	29
	2	liveme	24
	3	NoDAPL	22
	4	WaterIsLife	22
	5	nyc	20
🎃	1	halloween	367
	2	Halloween	325
	3	HappyHalloween	177
	4	happyhalloween	133
	5	fall	101
🎉	1	SoundHound	31
	2	happybirthday	20
	3	1	19
	4	birthday	18
	5	HappyBirthday	17
🎶	1	ExceptoAti_MS	56
	2	TengoQueColgar	56
	3	music	36
	4	Music	24
	5	np	20
👌	1	STARZFOLLOWTRAIN	132
	2	BORICUAAAA	41
	3	PRIMO	19
	4	MELAO	16
	5	ARIAS	11
👍	1	MAGA	18
	2	love	15
	3	OTLFP	15
	4	bodybuilding	12
	5	model	12

Brazil

t..	Hashtag	
1	love	60
2	NationalBoyfri..	25
3	tbt	20
4	family	17
5	Repost	17
1	BiduanCantik	13
2	BintangPantura	13
3	TeamJaGo	12
4	WonderfulDan..	9
5	DolanTwinsNe..	8
1	halloween	##
2	Halloween	76
3	いいねした人に..	39
4	HappyHallowe..	34
5	happyhalloween	28
1	LABEL	13
2	NowPlaying	8
3	birthday	7
4	party	6
5	happybirthday	6
1	RaphaelJabbour	37
2	MeBeija	15
3	QueroNoTVZ	14
4	music	10
5	AfueraEstaLlov..	7
1	STARZFOLLOW..	13
2	الديرجيست_سهلاويه	4
3	مجنون	4
4	BORICUAAAA	4
5	Repost	3
1	DolanTwinsNe..	12
2	NashsNewVideo	8
3	MatthewsNew..	6
4	INTELLECTUAL..	5
5	omgSG50	5

Japan

E..	t..	Hashtag	
❤	1	いいねした..	42
	2	デリヘル	29
	3	ハロウィン	27
	4	秋	26
	5	love	25
✨	1	占い	27
	2	ハロウィン	26
	3	japan	21
	4	いいねした..	21
	5	出会い	21
🎃	1	いいねした..	##
	2	ハロウィン	##
	3	halloween	##
	4	DMでやる	##
	5	Halloween	60
🎉	1	埼玉	12
	2	いいねした..	11
	3	高収入	10
	4	バイト	9
	5	おめでとう	6
🎶	1	埼玉	19
	2	NowPlaying	14
	3	高収入	14
	4	バイト	11
	5	ゼルビ屋	9
👌	1	いいねした..	8
	2	沖縄	5
	3	DMでやる	4
	4	いいねした..	4
	5	カラー	3
👍	1	いいねした..	51
	2	いいねした..	18
	3	DMでやる	7
	4	ハロウィン	6
	5	ラーメン	6

Mexico

E..	t..	Hashtag	
❤	1	UrbandLine	75
	2	AfueraEsta..	54
	3	TengoQueC..	43
	4	ElPerdedor	42
	5	MeFaltaUn..	37
✨	1	DistritoZio..	15
	2	OYELAZON..	15
	3	UrbandLine	15
	4	MatthewsN..	9
	5	cdmx	7
🎃	1	Halloween	14
	2	halloween	13
	3	Jaloguin	4
	4	Acabatelo	3
	5	CDMX	3
🎉	1	AfueraEsta..	21
	2	HBDJosCD9	7
	3	party	7
	4	Davalos5k	6
	5	AquiEstoyT..	5
🎶	1	AfueraEsta..	71
	2	MariasTrue..	16
	3	PorQueMe..	15
	4	MeFaltaUn..	13
	5	DesdeQueT..	7
👌	1	AfueraEsta..	22
	2	UrbandLine	15
	3	FxTeam	9
	4	AbrahamM..	5
	5	AskJefeDeJ..	5
👍	1	AWLMusicV..	11
	2	MexicoGP	5
	3	AcademiaV..	3
	4	F1	3
	5	CDMX	2

United Kingdom

E..	t..	Hashtag	
❤	1	BiduanCantik	83
	2	WonderfulDangdut	81
	3	BintangPantura	80
	4	TeamJaGo	74
	5	love	49
✨	1	BiduanCantik	83
	2	WonderfulDangdut	81
	3	BintangPantura	80
	4	TeamJaGo	74
	5	DolanTwinsNewVideo	9
🎃	1	halloween	##
	2	Halloween	##
	3	HappyHalloween	61
	4	pumpkin	40
	5	happyhalloween	34
🎉	1	NowPlaying	82
	2	ClubLife	18
	3	HouseMusic	13
	4	Rewind90s	13
	5	birthday	11
🎶	1	London	6
	2	excited	4
	3	Joel	4
	4	music	4
	5	countrymusic	3
👌	1	XFactor	9
	2	london	8
	3	AYE	4
	4	LFC	4
	5	Repost	4
👍	1	scarborough	8
	2	ukrunchat	7
	3	StandUpToCancer	6
	4	win	6
	5	awesome	5

Fig. 27. Top-5 hashtag counts for popular emojis, disaggregated by country (for a sample of 5 countries).

its relative popularity compared to other emojis can depend on the particular country. In Brazil, for example, the ❤️ emoji is (slightly) more popular. In the Philippines and Myanmar, this emoji is far less popular (especially in the latter). Therefore, while one or two emojis would seem to be universally popular, even for these emojis, the relative popularity can differ depending on the country. A similar qualitative study applies for languages (Fig. 22).

Although our findings thus far have been based on analyzing data from an aggregate or distributional standpoint, it is also instructive, especially given sociological interest in this subject, to conduct specific case studies either with respect to a language or a country. Given that our corpus is geotagged, accurate visualizations can be produced. Fig. 23 provides a global view of tweets in English that contain at least one emoji. Given that there is such geographical dispersion or 'spread' in the use of English for tweeting, some of the conclusions in Fig. 13 start to make more sense. In that figure, English has moderately high divergence from other languages, but not too much. In view of this qualitative finding, one suggestion is that language-dependence on emoji usage may be partially explainable by the spread of the language. Languages that are used in a highly localized region (which could be an entire country, such as Japan) may then be hypothesized to have high divergence. However, it is unlikely that this is the only explanation, or even an explanation that is universally valid. More sociological studies are needed to understand the effects of spread on the use of non-linguistic, but culturally relevant, symbols of digital communication such as emojis. For instance, linguistic spread may not matter as much once historical background (such as the Anglo-colonial backgrounds shared by many of the commonwealth nations today) is accounted for. Regression-based methods would be appropriate for understanding which variables have both significant and positive effects.

We can do a similar 'case study' analysis for a single country or region, rather than a language. Fig. 24 shows a regional plot (the US and Mexico). Unsurprisingly, English dominates in this region, though Spanish becomes much more prevalent (again, not surprisingly) in Mexico. Arabic seems to be the third-most prevalent language, and primarily focused in the more densely populated and diverse regions of the US (the West Coast, the Northeast, some pockets of the state of Texas, and the eastern coastline of the state of Florida). Other languages are present, but in such minor concentrations, that they are almost not visible. Although a full sociological analysis of such case studies is not within the scope of this work, researchers wishing to do so could potentially use our dataset both for contextualizing some of our earlier findings, and for doing comparative analyzes for specific countries, regions and languages. Additionally, it may be an interesting exercise to replicate some of these findings using data from other social media sources, such as Facebook, Snapchat and Instagram. Some findings will likely be found to hold only for Twitter, while others may be more general and universal. Discovering these more universal variables is a valuable area for further research, as is quantifying their relative effects on emoji usage and online discourse.

Finally, we briefly discuss how the findings herein could be applied to improving sentiment analysis and opinion mining algorithms, especially in multi-lingual and culturally diverse contexts. One avenue is to not treat emojis as universal 'words' that have the same implication in every language and country, but to instead assign it a 'prior' score based on how often (and in what normative conditions) that emoji is used in the context of that language or country. For example, the 'heart' emoji was found to be much more common in some of the Arabic-speaking Middle-Eastern nations than in Western nations that tended to favor smiley-based emojis more. A heart emoji would have a different prior in some nations than in others. These priors could be incorporated as features in any machine learning-based sentiment analysis algorithm that is trained for that language or country. Using

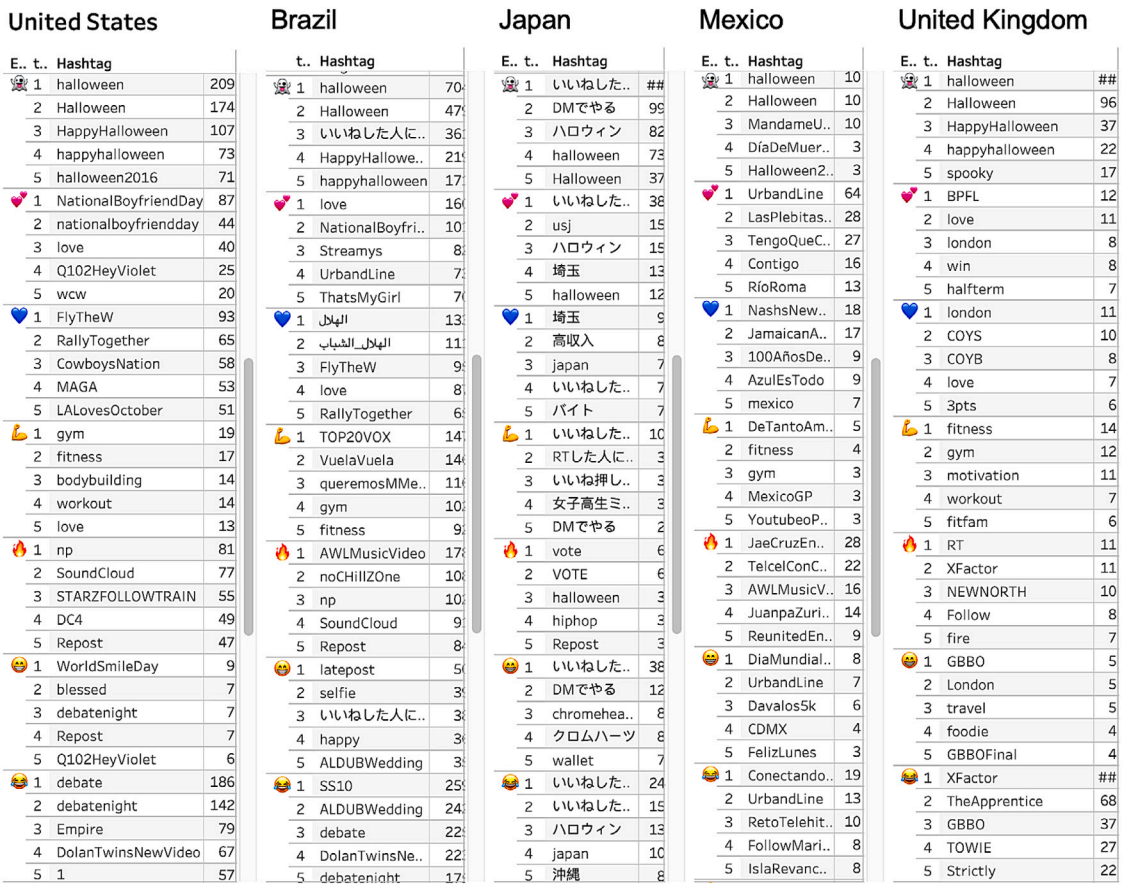


Fig. 28. Top-5 hashtag counts for popular emojis, disaggregated by country (for a sample of 5 countries).

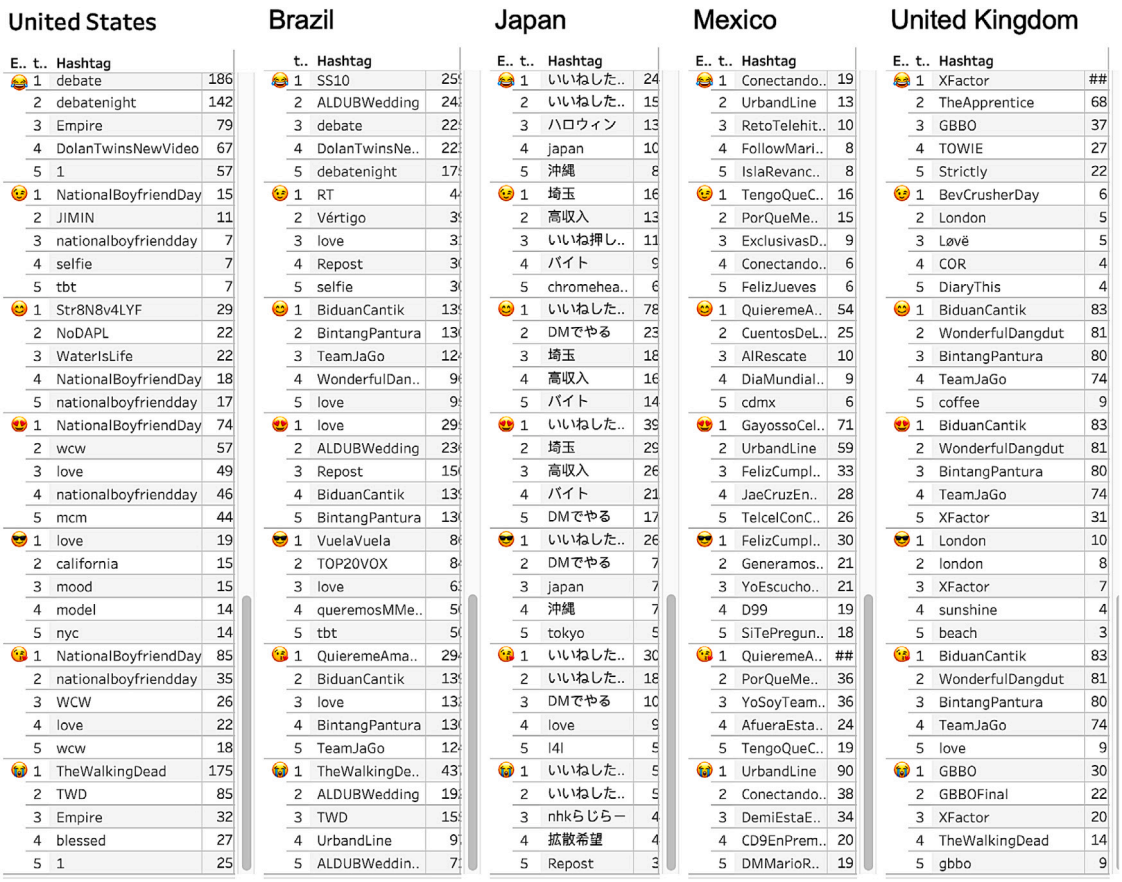
techniques such as parameter sharing, and by explicitly incorporating interaction effects between emoji, language and country, a globally applicable sentiment analysis model may yield higher accuracy than what is currently possible. This paper showed that such interaction effects are indeed meaningful and significant. Similar to how interaction effects can sometimes help yield regression models with better goodness-of-fit (R^2), interaction effects between emoji, language and country, encoded as features, may help yield much higher-quality (and potentially, more *explainable*) sentiment analysis models.

We end this section by re-iterating that all results reported in this paper must be interpreted in two contexts. First, since we only considered tweets where users enabled geolocation (and for which explicit location information is therefore available), we cannot necessarily extend these results to tweets where this information is not available, as other work has shown that tweets with geolocations may be subject to their own biases [35]. Hence, this is an important area for future work. Second, since the data is from 2016, it is also not necessarily true that we can extend our results to 2021. However, since these results can serve as a reference, the question of how, and to what extent, recent emoji usage has changed since then could also serve as a valuable avenue for future research. Furthermore, by treating the set of geolocation-enabled tweets as a 'target group', and the set of other (non-geolocation-enabled) tweets as a 'control group', and replicating and comparing many of the results herein between these two groups, we may learn more about how the particular feature of geolocation-enabling impacts emoji usage and emotive expressiveness on social media.

6. Future work and conclusion

Emojis have emerged as ubiquitous features of digital communication. They are used widely in text messages as well as on all major social media platforms. Despite their ubiquity in human social media communication, there are many questions about emoji usage that had not been addressed in detail (and at scale) prior to this work, including linguistic and country correlates, as well as the differences in emoji usage across languages and countries. In this article, we designed a detailed empirical study for answering some of these questions. We found that there are strong dependencies at both the language and country level, with the latter causing more variance in emoji usage distributions than the former. Some languages and countries are also found to be much more constrained in the diversity of their emoji usage. Similarly, information-theoretic results show that some languages and countries are more divergent in their emoji usage distributions than others, compared to the distributions of all other languages. Because emojis have a cultural dimension, we suspect that such observed similarities and dissimilarities may have (among other causes) a cultural basis.

Future studies could consider supplementing these studies by considering emoji multi-networks and weighted networks to study issues such as emoji co-occurrence, which we only briefly discussed in Section 5. Previous preliminary studies on this matter have illustrated some interesting trends [24,25,42], but much more research is possible by using the full suite of network science tools. For example, one promising avenue that we did not cover in this article but that might shed more insight into emoji usage is *hashtag–emoji* co-occurrence. Specifically, are hashtags substitutes to, or complements for, emojis? In very early experiments that we conducted, we found that hashtags and emojis co-occur surprisingly less often than might be expected (less than 5% in our corpus). Therefore, either hashtags and emojis are 'competing' for consideration within a tweet, or the two serve very distinct purposes. Emojis may be used more often for emotional expression, while hashtags may indicate shared topical interests. Studying their intersection, as well as other independent variables such as language



United States

E..	t..	Hashtag	
😂	1	debate	186
	2	debatenight	142
	3	Empire	79
	4	DolanTwinsNewVideo	67
	5	1	57
😉	1	NationalBoyfriendDay	15
	2	JIMIN	11
	3	nationalboyfriendday	7
	4	selfie	7
	5	tbt	7
😊	1	Str8N8v4LYF	29
	2	NoDAPL	22
	3	WaterIsLife	22
	4	NationalBoyfriendDay	18
	5	nationalboyfriendday	17
😍	1	NationalBoyfriendDay	74
	2	wcw	57
	3	love	49
	4	nationalboyfriendday	46
	5	mcm	44
😎	1	love	19
	2	california	15
	3	mood	15
	4	model	14
	5	nyc	14
😘	1	NationalBoyfriendDay	85
	2	nationalboyfriendday	35
	3	WCW	26
	4	love	22
	5	wcw	18
😡	1	TheWalkingDead	175
	2	TWD	85
	3	Empire	32
	4	blessed	27
	5	1	25

Brazil

t..	Hashtag	
😂 1	SS10	25
2	ALDUBWedding	24
3	debate	22
4	DolanTwinsNe..	22
5	debatenight	17
😉 1	RT	4
2	Vértigo	3
3	love	3
4	Repost	3
5	selfie	3
😊 1	BiduanCantik	13
2	BintangPantura	13
3	TeamJaGo	12
4	WonderfulDan..	9
5	love	9
😍 1	love	29
2	ALDUBWedding	23
3	Repost	15
4	BiduanCantik	13
5	BintangPantura	13
😎 1	VuelaVuela	8
2	TOP20VOX	8
3	love	6
4	queremosMMe..	5
5	tbt	5
😘 1	QuieremeAma..	29
2	BiduanCantik	13
3	love	13
4	BintangPantura	13
5	TeamJaGo	12
😡 1	TheWalkingDe..	43
2	ALDUBWedding	19
3	TWD	15
4	UrbandLine	9
5	ALDUBWeddin..	7

Japan

E..	t..	Hashtag	
😂	1	いいねした..	24
	2	いいねした..	15
	3	ハロウィン	13
	4	japan	10
	5	沖縄	8
😉	1	埼玉	16
	2	高収入	13
	3	いいね押し..	11
	4	バイト	9
	5	chromehea..	6
😊	1	いいねした..	78
	2	DMでやる	23
	3	埼玉	18
	4	高収入	16
	5	バイト	14
😍	1	いいねした..	39
	2	埼玉	29
	3	高収入	26
	4	バイト	21
	5	DMでやる	17
😎	1	いいねした..	26
	2	DMでやる	7
	3	japan	7
	4	沖縄	7
	5	tokyo	5
😘	1	いいねした..	30
	2	いいねした..	18
	3	DMでやる	10
	4	love	9
	5	l4l	5
😡	1	いいねした..	5
	2	いいねした..	5
	3	nhkらじらー	4
	4	拡散希望	4
	5	Repost	3

Mexico

E..	t..	Hashtag	
😂	1	Conectando..	19
	2	UrbandLine	13
	3	RetoTelehit..	10
	4	FollowMari..	8
	5	IslaRevanc..	8
😉	1	TengoQueC..	16
	2	PorQueMe..	15
	3	ExclusivasD..	9
	4	Conectando..	6
	5	FelizJueves	6
😊	1	QuieremeA..	54
	2	CuentosDeL..	25
	3	AlRescate	10
	4	DiaMundial..	9
	5	cdmx	6
😍	1	GayossoCel..	71
	2	UrbandLine	59
	3	FelizCumpl..	33
	4	JaeCruzEn..	28
	5	TelcelConC..	26
😎	1	FelizCumpl..	30
	2	Generamos..	21
	3	YoEscucho..	21
	4	D99	19
	5	SiTePregun..	18
😘	1	QuieremeA..	##
	2	PorQueMe..	36
	3	YoSoyTeam..	36
	4	AfueraEsta..	24
	5	TengoQueC..	19
😡	1	UrbandLine	90
	2	Conectando..	38
	3	DemiEstaE..	34
	4	CD9EnPrem..	20
	5	DMMarioR..	19

United Kingdom

E..	t..	Hashtag	
😂	1	XFactor	##
	2	TheApprentice	68
	3	GBBO	37
	4	TOWIE	27
	5	Strictly	22
😉	1	BevCrusherDay	6
	2	London	5
	3	Løvë	5
	4	COR	4
	5	DiaryThis	4
😊	1	BiduanCantik	83
	2	WonderfulDangdut	81
	3	BintangPantura	80
	4	TeamJaGo	74
	5	coffee	9
😍	1	BiduanCantik	83
	2	WonderfulDangdut	81
	3	BintangPantura	80
	4	TeamJaGo	74
	5	XFactor	31
😎	1	London	10
	2	london	8
	3	XFactor	7
	4	sunshine	4
	5	beach	3
😘	1	BiduanCantik	83
	2	WonderfulDangdut	81
	3	BintangPantura	80
	4	TeamJaGo	74
	5	love	9
😡	1	GBBO	30
	2	GBBOFinal	22
	3	XFactor	20
	4	TheWalkingDead	14
	5	gbbo	9

Fig. 29. Top-5 hashtag counts for popular emojis, disaggregated by country (for a sample of 5 countries).

and country that may serve as further influences on intersectional properties, may provide valuable insights into how these units of discourse (that are specific to modern-day digital communications) affect, and are affected by, users' shared linguistic and cultural milieus. We provide more details, and early results, in the appendix, on a select set of intersectional properties between hashtags and popular emojis, but far more study is necessary before the interaction effects of emoji and hashtag usage can be fully understood. Furthermore, these effects will likely differ based on whether geolocation is enabled or not.

A related, but different, line of study that may further reveal deep-seated linguistic and cultural differences (especially the latter) is the *tone* of the tweets in which popular emojis are used. For example, we may learn that one country's emoji tweets are overly personal, while those from another country are more political. These are, of course, correlated with the cultural and political norms in the country, but thus far, there have not been many reliable ways of measuring and quantifying such norms using 'grassroots' data such as social media. Further studies on this issue may provide novel ways of conducting such measurements in a cost-effective manner, especially as a younger, more vocal generation comes online and Internet connectivity (and smartphone usage) continues to improve across the globe.

CRediT authorship contribution statement

Mayank Kejriwal: Conceptualization, Supervision, Writing - review & editing, Writing - original draft, Methodology. **Qile Wang:** Data curation, Software, Visualization, Investigation. **Hongyu Li:** Data curation, Software, Visualization, Investigation, Methodology. **Lu Wang:** Data curation, Software, Visualization, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Language and country codes

Table 1 below provide expansions for the language and country codes used in this paper.

Tables 2 and 3 provide the number of tweets (and also the number of tweets with emojis, expressed also as a percentage over the total number of tweets), demarcated respectively by language and country, present in our corpus and used in this paper.

Appendix B. Additional details on the 3-phase structure

Fig. 25 provides a breakdown by category for each of the three phases that were identified earlier in the paper in Fig. 6. To assign an emoji to a category, we used a publicly available Web resource.[11]

Appendix C. Some preliminary results on hashtag–emoji occurrence

Fig. 26 show the counts of the top-10 hashtags for a sample of popular emojis. Some hashtags are predictable in the context of the emoji: for example, for the 'heart' emoji at the top-left of Fig. 26, we find hashtags such as 'love', 'family' and 'friends'. We also see that the

[11] https://www.webfx.com/tools/emoji-cheat-sheet/.

Table 1
Expansions for the language and country codes used in this paper.

	Language code		Country code	
1	**en**	English	**US**	United States
2	**pt**	Portuguese	**BR**	Brazil
3	**es**	Spanish	**JP**	Japan
4	**ja**	Japanese	**GB**	Great Britain (UK)
5	**in**	Former Indonesian	**AR**	Argentina
6	**ar**	Arabic	**PH**	Philippines
7	**tr**	Turkish	**MY**	Malaysia
8	**tl**	Tagalog	**TR**	Turkey
9	**fr**	French	**ES**	Spain
10	**ru**	Russian	**FR**	France
11	**th**	Thai	**ID**	Indonesia
12	**it**	Italian	**MX**	Mexico
13	**nl**	Dutch	**TH**	Thailand
14	**de**	German	**SA**	Saudi Arabia
15	**pl**	Polish	**CA**	Canada
16	**ht**	Ahtena	**RU**	Russian Federation
17	**ko**	Korean	**CO**	Colombia
18	**et**	Estonian	**IT**	Italy
19	**sv**	Swedish	**IN**	India
20	**hi**	Hindi	**VE**	Venezuela
21	**fi**	Finnish	**PT**	Portugal
22	**zh**	Chinese	**ZA**	South Africa
23	**da**	Danish	**CL**	Chile
24	**eu**	Basque	**AU**	Australia
25	**no**	Norwegian	**DE**	Germany
26	**ro**	Romanian	**EG**	Egypt
27	**cs**	Czech	**UY**	Uruguay
28	**cy**	Welch	**KW**	Kuwait
29	**el**	Greek	**NL**	Netherlands
30	**uk**	Ukrainian	**AE**	United Arab Emirates

Table 2
Tweet prevalence in our corpus by language.

Language	Total number of tweets	Number of tweets with emojis	% of tweets with emojis
en	86,59,884	19,76,783	22.83%
pt	23,00,832	3,69,045	16.04%
es	21,44,723	5,22,167	24.35%
ja	11,02,047	2,13,238	19.35%
in	6,93,971	2,27,285	32.75%
ar	5,38,868	2,01,167	37.33%
tr	4,54,221	75,421	16.60%
tl	4,12,892	1,38,713	33.60%
fr	3,92,543	82,992	21.14%
ru	2,93,473	60,869	20.74%
th	2,73,437	44,822	16.39%
it	1,62,638	30,119	18.52%
nl	90,274	14,171	15.70%
de	84,577	14,934	17.66%
pl	53,522	8133	15.20%
ht	50,250	11,237	22.36%
ko	41,611	3523	8.47%
et	37,647	8302	22.05%
sv	36,556	6163	16.86%
hi	32,989	5665	17.17%
fi	26,102	2974	11.39%
zh	16,324	3099	18.98%
da	15,943	2987	18.74%
eu	15,067	3433	22.78%
no	14,836	2677	18.04%
ro	13,560	2587	19.08%
cs	12,788	2944	23.02%
cy	12,194	3080	25.26%
el	12,070	880	7.29%
uk	12,064	2243	18.59%

Table 3
Tweet prevalence in our corpus by country.

Country	Total number of tweets	Number of tweets with emojis	% of tweets with emojis
US	67,26,826	14,05,940	20.90%
BR	24,99,041	3,70,903	14.84%
JP	11,24,317	2,14,580	19.09%
GB	9,67,584	2,08,275	21.53%
AR	7,48,262	1,97,622	26.41%
PH	5,79,842	2,07,716	35.82%
MY	5,07,092	1,52,772	30.13%
TR	4,98,124	75,642	15.19%
ES	4,75,227	88,890	18.70%
FR	4,04,235	78,362	19.39%
ID	3,74,214	1,36,733	36.54%
MX	3,46,972	93,773	27.03%
TH	3,39,530	50,748	14.95%
SA	2,82,744	83,936	29.69%
CA	2,79,528	37,297	13.34%
RU	2,48,574	48,654	19.57%
CO	1,82,268	37,402	20.52%
IT	1,79,394	30,647	17.08%
IN	1,78,464	20,579	11.53%
VE	1,45,321	11,747	8.08%
PT	1,36,169	17,512	12.86%
ZA	1,35,426	37,239	27.50%
CL	1,21,007	24,865	20.55%
AU	1,19,853	18,352	15.31%
DE	1,11,564	17,204	15.42%
EG	1,00,959	34,749	34.42%
UY	1,00,386	20,131	20.05%
KW	1,00,063	35,410	35.39%
NL	92,981	11,412	12.27%
AE	84,995	29,866	35.14%

most common hashtags for the blue-heart emoji are in Arabic, since this particular emoji is very popular in many of the Arab-speaking Middle-Eastern nations. In contrast, some other hashtags seem to have event-dependencies. The most obvious is the pumpkin emoji, associated with halloween-related hashtags, but other emojis (including the 'like' or thumbs-up emoji) are associated with events such as 'AWLMusicVideo' and 'MatthewsNewVid'. The double-heart emoji in the bottom left of Fig. 26 is associated with 'NationalBoyfriendDay' and 'ThatsMyGirl'. Clearly, there is a lot to be learned from the intersection of hashtags and emojis; we leave for future work to study this intersectionality as intuited in Section 6.

Figs. 27, 28 and 29 illustrate similar results, but broken down by country, for a sample of 5 countries that were found to use emojis significantly. Language-specific differences immediately stand out, especially involving hashtags tweeted in Japan and Mexico. Interestingly, there are many English hashtags used in Brazil. Country-specific differences are also apparent; for example, in looking at the top hashtags in the United States versus the United Kingdom for the 'like' hashtag at the bottom of Fig. 27, we find country-specific hashtags like 'MAGA' and 'ukrunchat' for the two countries, respectively. Culture-specific differences are also observable e.g., 'Empire' versus 'XFactor' in the US versus the UK, respectively, for the hashtag at the bottom of Fig. 28. These findings show that, even (or especially) when studying hashtag–emoji intersectionality, country, language and cultural dependencies, may all have to be taken into account in a full sociological treatment.

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